

Fair-Value Capital and Bank Solvency over the Monetary Policy Cycle*

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July 2026

Abstract

We develop a quantitative general equilibrium model of banks that issue credit-risky loans and invest in interest rate-sensitive securities to study how including unrealized securities gains and losses in regulatory capital affects lending and solvency over the monetary policy cycle. Regulatory accounting rules generate asymmetric effects across the cycle. During tightening, including valuation changes in regulatory capital substantially reduces bank failures while only moderately contracting credit. During easing, it substantially expands credit while only moderately increasing failures. The asymmetry arises from the interaction of credit and interest rate risk. Welfare gains from incorporating valuation changes are modest.

Keywords: Interest rate risk, credit risk, regulatory capital, accounting rules, financial stability

JEL Classification: G21, G28, G32, G33, E44, E51

*We are extremely grateful to Rafael Repullo and Javier Suarez for inspiration, guidance and support with this project. We have also benefited from helpful comments and suggestions provided by Amil Dasgupta, Tetiana Davydiuk, David Martinez-Miera, Lorian Pelizzon, B Ravikumar and Anatoli Segura. Part of this work was completed during Agarwal's PhD studies at CEMFI. Schäfer acknowledges financial support from grant PRE2022-101467, the Maria de Maeztu Unit of Excellence CEMFI MDM-2016-0684, funded by MCIN/AEI/10.13039/501100011033, and CEMFI.

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1 Introduction

Banks invest in large portfolios of long-term fixed-income securities that are highly liquid and nearly free of default risk. These securities allow banks to meet unexpected withdrawals from their predominantly short-term funding while earning a term premium, but also expose them to interest rate risk. When interest rates rise, the market value of long-term securities declines, reducing the economic value of bank equity and potentially constraining lending. Financial regulators have historically mitigated these effects through prudential filters that partially exclude unrealized gains and losses on securities portfolios from regulatory capital calculations.¹

The main concern with excluding unrealized gains and losses on securities is that doing so may weaken the ability of regulatory capital to reflect banks’ true loss-absorbing capacity (Laux, 2012). These concerns became salient during the 2023 banking turmoil, when sharp increases in interest rates generated large unrealized losses on banks’ long-term securities portfolios. By excluding these losses from regulatory capital, prudential filters allowed a large set of banks to operate with elevated effective leverage (Flannery and Sorescu, 2023), reducing their capacity to absorb subsequent losses and contributing to the failures of Silicon Valley Bank and First Republic. These events have prompted a debate on the optimal regulatory treatment of banks’ securities portfolios (Basel Committee on Banking Supervision, 2023).

We contribute to these discussions using a quantitative macro-banking model with two key features. First, it distinguishes between the fair value of bank equity, which incorporates unrealized gains and losses arising from interest rate fluctuations, and the book value of equity, which does not. Second, it jointly analyzes banks’ exposure to credit and interest rate risk.² We use the model to evaluate how capital regulation based on the book or fair value of bank equity affects lending and solvency over the monetary policy cycle.

¹For the historic treatment, see for instance the *Guidelines on Prudential Filters for Regulatory Capital* (December 21, 2004) published by the Committee of European Banking Supervisors (CEBS): <https://www.eba.europa.eu/guidelines-2004>; and the arguments presented in the U.S. FDIC’s *Final Rule on Regulatory Capital Treatment of Unrealized Holding Gains and Losses Under FASB 115* (December 28, 1994): <https://www.fdic.gov/news/inactive-financial-institution-letters/1995/fil9503a.html>. See Argimón, Dietsch, and Estrada (2018) and Kim, Kim, and Ryan (2019) for overviews of prudential filters that exclude unrealized gains and losses from regulatory capital across euro area countries and in the U.S., respectively.

²The importance of jointly accounting for both risks in modeling banks has been emphasized by Breuer, Jandačka, Rheinberger, and Summer (2010), Drehmann, Sorensen, and Stringa (2010), and Begenau, Piazzi, and Schneider (2025b), among others.

In our quantitative framework, banks extend one-period risky loans to entrepreneurial firms and invest in default-free long-term government bonds using insured deposits and equity. As in [Mendicino, Nikolov, Rubio Ramírez, Suarez, and Supera \(2026\)](#), the risk of firm default is not fully diversifiable at the bank level, implying that bank solvency problems arise endogenously from high default rates among bank borrowers. Investment in long-term bonds creates a maturity mismatch that exposes banks to interest rate risk. We embed this framework in a standard New-Keynesian model with investment. Banks also operate a deposit franchise, paying deposit rates that are low and relatively insensitive to the policy rate, consistent with the stable net interest margins documented by [Drechsler, Savov, and Schnabl \(2021\)](#). This mechanism partially hedges, but does not eliminate, the interest rate risk associated with long-term bond holdings.

Banks are subject to risk-based capital requirements that assign positive risk weights to risky loans and zero risk weights to long-term government bonds. As in existing macro-banking models, capital regulation limits the excessive risk-taking incentives created by limited liability and safety net guarantees. A key feature of our framework is that regulatory constraints are based on the book value of equity, while bank solvency depends on the economic value of assets and liabilities. Consequently, unrealized gains and losses on long-term securities affect banks' economic net worth but may be excluded from regulatory capital. During monetary tightening episodes, declines in bond prices therefore allow banks to operate with elevated effective leverage relative to their true loss-absorbing capacity, amplifying the effects of borrower defaults on bank solvency and macroeconomic outcomes.

To discipline the quantitative analysis, we estimate the model using the simulated method of moments, targeting macroeconomic, banking, and financial moments from euro area data over the period 1995–2016. The estimated model replicates key features of bank balance sheets, including the average loans-to-bonds ratio, the maturity structure of bond holdings, and the sensitivity of the deposit rate to the policy rate (i.e., the deposit beta) that govern banks' exposure to interest rate risk. It also generates loan pricing responses consistent with empirical evidence: banks raise lending rates following unrealized losses on their bond portfolio, although less strongly than following realized reductions in regulatory capital. Finally, the model produces episodes of large unrealized losses associated with sharp monetary policy tightening, akin to the historical experience during the post-COVID monetary tightening cycle ([Marsh and Laliberte, 2023](#)).

Our main exercise studies alternative regulatory accounting rules in this environment. We compare a book-value-based capital requirement with a fair value-based requirement that incorporates unrealized gains and losses on banks' long-term bond portfolios into regulatory capital. Recognizing these valuation changes makes regulatory capital more sensitive to interest rate movements and therefore increases the volatility of credit supply when capital requirements bind. At the same time, it better aligns regulatory constraints with banks' true loss-absorbing capacity. When interest rates rise and losses accumulate on bond portfolios, banks become more constrained, charge higher loan spreads, and operate with lower effective leverage, reducing their probability of default. When interest rates fall, valuation gains loosen regulatory constraints, lower loan spreads, and permit banks to operate with higher effective leverage.

The effects on bank solvency are highly asymmetric over the monetary policy cycle. The asymmetry arises because realizations of credit risk and interest rate risk tend to co-move over the cycle. During tightening episodes, rising interest rates simultaneously depress bond values and increase borrower default risk, making banks particularly vulnerable. By constraining leverage during these periods, a fair value-based capital requirement substantially reduces bank default risk. During easing episodes, by contrast, the additional leverage capacity created by valuation gains leads to only a moderate increase in bank default risk because lower interest rates also reduce borrower defaults. In the estimated model, the reduction in bank default probability during tightening is approximately thirty times larger than the increase during easing.

This asymmetric effect on bank solvency, in turn, shapes the response of aggregate credit supply. Under book-value-based capital requirements, the relatively looser lending constraints faced by banks during monetary tightening are largely offset by higher bank failures and the resulting losses in aggregate bank capital. Fair value-based requirements therefore produce only a moderately larger contraction in credit, despite tightening regulatory constraints during these episodes. During easing episodes, the additional lending capacity created by recognition of valuation gains under fair-value requirements is only partially offset by the modest increase in bank failures, leading to a substantial expansion in credit. Quantitatively, the credit expansion during easing is approximately twice as large as the contraction during tightening.

These asymmetric effects lower bank and firm default probabilities on average, reducing

the deadweight costs associated with bankruptcy and freeing resources for consumption. Although fair value-based capital requirements increase the volatility of credit, they raise average consumption and reduce consumption volatility. The resulting welfare gains are modest.

Our findings are consistent with the rationale for regulatory frameworks that include unrealized gains and losses on banks’ available-for-sale (AFS) securities in Tier 1 capital, as under Basel III ([Basel Committee on Banking Supervision, 2019](#), CAP10.6(4)). They also suggest that extending similar recognition to held-to-maturity (HTM) securities could improve welfare. From a modeling perspective, our findings reiterate the importance of jointly analyzing banks’ credit and interest rate risk exposures ([Begenau et al., 2025b](#)).

Related literature This paper contributes to the literature on quantitative models of bank regulation. Following the formalization of the welfare trade-offs associated with capital regulation by [Van den Heuvel \(2008\)](#), a large literature studies the optimal level of capital requirements ([Begenau, 2020](#); [Elenev, Landvoigt, and Van Nieuwerburgh, 2021](#); [Abad, Martinez-Miera, and Suarez, 2024](#); [Mendicino et al., 2026](#)), their interaction with monetary policy ([Angeloni and Faia, 2013](#); [Collard, Dellas, Diba, and Loisel, 2017](#)) and banking market structure ([Corbae and D’Erasmus, 2021](#); [Begenau and Landvoigt, 2022](#)), and the rationale for time-varying requirements ([Davydiuk, 2017](#); [Malherbe, 2020](#)). In most of these models, bank assets and liabilities mature within one period, abstracting from the interest rate risk generated by maturity mismatch. We introduce long-term government bonds into banks’ portfolio choice problem and study how the regulatory accounting treatment of their valuation changes interacts with capital regulation, credit supply, and bank solvency.

Among quantitative models of bank regulation, our paper is most closely related to [Begenau, Bigio, Majerovitz, and Vieyra \(2025a\)](#), in which book and fair-valued bank equity differ because accounting rules permit delayed recognition of loan losses. Requiring instantaneous recognition of loan losses effectively tightens capital requirements, reducing bank defaults at the cost of lower credit supply.³ In our model, by contrast, the recognition of unrealized gains and losses on long-term bonds can either tighten or relax regulatory constraints as in-

³[Abad, Ikeda, and Suarez \(2025\)](#) compare incurred-loss provisioning with the expected credit loss approaches of IFRS 9 and CECL. They show that earlier recognition of expected loan losses raises the equity funding needed to support lending, reducing its average level and increasing its cyclicalit.

terest rates change. This state-dependent effect and its interaction with credit risk generates asymmetric responses of bank solvency and credit supply over the monetary policy cycle.

Our work is also related to the literature on accounting rules and the measurement of bank capital.⁴ A prominent argument against fair-value accounting is that using market prices to assess the solvency of financial institutions may lead to inefficient failures when markets are illiquid, as prices can deviate from fundamental values during periods of financial stress (see, e.g., [Allen and Carletti, 2008](#); [Plantin, Sapra, and Shin, 2008](#)).⁵ When financial institutions are subject to capital requirements, marking assets to distressed prices can deplete regulatory capital and induce forced sales, potentially generating fire-sale price spirals ([Laux and Leuz, 2010](#); [Ellul, Jotikasthira, and Lundblad, 2011](#)).⁶ [Ellul, Jotikasthira, Lundblad, and Wang \(2015\)](#) provide evidence that, during the financial crisis, historical-cost accounting induced gains trading: insurers delayed selling downgraded risky assets and instead realized gains by selling otherwise unrelated securities. [Acharya, Mukherjee, and Sundaram \(2021\)](#) provide an ex ante welfare analysis of fair-value and historical-cost accounting in which fair-value disclosure improves financing and project-continuation decisions but may also trigger creditor runs, fire sales, and contagion. Our framework differs from these papers because the assets subject to revaluation are liquid, default-free government bonds whose prices respond primarily to interest-rate movements and which receive a zero regulatory risk weight in many jurisdictions.

The treatment of unrealized gains and losses on such securities in regulatory capital became a central policy issue following the 2023 U.S. banking turmoil (see [Barr, 2023](#)). [Granja, Jiang, Matvos, Piskorski, and Seru \(2024\)](#) document that banks relied on held-to-maturity accounting, rather than economic hedging, to shield book capital from rising-rate losses, and study how accounting and capital regulation shape hedging and recapitalization incentives. [Greenwald, Krainer, and Paul \(2024\)](#) and [Orame, Ramcharan, and Robatto \(2025\)](#) study how regulatory accounting affects the transmission of monetary policy to bank lending, showing that lending responds more strongly to monetary policy under fair-value account-

⁴See [Freixas and Tsomocos \(2004\)](#) for an early theoretical contribution.

⁵[Heaton, Lucas, and McDonald \(2010\)](#) study the interaction between accounting rules and regulatory capital requirements and show that fair-value accounting can create inefficiencies even when market prices reflect fundamental values.

⁶[Laux and Leuz \(2010\)](#) conclude, however, that fair-value accounting was unlikely to have contributed substantially to the severity of the 2008–2009 financial crisis.

ing.⁷ The lending implications of our model are consistent with these and related empirical findings (see e.g. [Beutler, Bichsel, Bruhin, and Danton, 2020](#); [Marsh and Laliberte, 2023](#)). We complement this literature by studying the financial-stability consequences of regulatory accounting and thereby connect it to recent work on the importance and implications of banks’ exposure to interest-rate risk ([Begenau et al., 2025b](#); [Drechsler et al., 2021](#); [Jiang, Matvos, Piskorski, and Seru, 2024](#); [DeMarzo, Krishnamurthy, and Nagel, 2024](#); [Begenau, Landvoigt, and Elenev, 2024](#)).

Outline The rest of the paper is organized as follows. Section 2 describes our macro-banking model. Section 3 contains the solution method and calibration strategy, presents the baseline parameterization, and discusses the quantitative performance of the model. In Section 4 we analyze the performance of the economy under alternative regulatory accounting frameworks, identifying the approach that maximizes social welfare. The Appendix contains a complete list of equilibrium conditions and full description of the data sources, solution method, and several complementary materials referred throughout the main text.

2 The Model

We consider a discrete-time, infinite-horizon economy in which dates are indexed by t . The baseline framework is a financial accelerator model embedded in a standard New Keynesian model with investment.⁸ Following the tradition of [Gertler and Karadi \(2011\)](#) each household consists of workers, entrepreneurs, and bankers. Workers supply labor to the production sector and transfer their wage income back to the household. Entrepreneurs and bankers provide equity to entrepreneurial firms and banks, respectively.

There exist a continuum of measure one of islands. In each island there is a continuum of measure one of entrepreneurial firms and a representative bank. Entrepreneurial firms and banks live for one period, issue equities among entrepreneurs and bankers, respectively, and obtain external financing by issuing non-contingent debt in the form of bank loans and deposits, respectively. Entrepreneurial firms use equity and loans to buy physical capital,

⁷Relatedly, [Gomez, Landier, Sraer, and Thesmar \(2021\)](#) find that banks experiencing increases (decreases) in net interest income following changes in the monetary policy rate expand (contract) lending relative to their peers.

⁸See e.g., [Smets and Wouters \(2007\)](#) and [Christiano, Eichenbaum, and Evans \(2005\)](#).

which some intermediate good producers rent in the next period. Their terminal net worth is subject to both idiosyncratic and island-specific shocks. The latter is non-diversifiable from the banks' perspective. In addition to providing loans, banks invest in a portfolio of long-term bonds. Both entrepreneurial firms and banks operate under limited liability and default when their terminal asset value is lower than their debt obligations. Non-defaulted entrepreneurial firms and banks pay their terminal net worth to entrepreneurs and bankers, respectively.

In the rest of this section, we present the model ingredients in more detail.

2.1 Households

There is a unit continuum of households indexed by h , that provide consumption insurance to three types of members: workers, bankers and entrepreneurs.

Households derive utility from consumption C_t and disutility from labor H_t . To improve the quantitative performance of the model, consumption is subject to internal habit formation governed by the parameter b .⁹ Households provide differentiated labor hours H_{ht} to intermediate goods-producing firms, remunerated at a nominal wage W_{ht} . The disutility derived from labor is governed by the inverse Frisch elasticity φ_H and a scaling parameter ξ_H .

Households can save in fully insured bank deposits D_t remunerated at gross interest rate R_{Dt} . To account for non-bank funding, households can also invest in physical capital K_t^H at real price Q_t , subject to a management cost ς_t , and rent it to intermediate good producers at rate z_t .¹⁰ Physical capital depreciates at rate δ . The nominal price of the single consumption good is denoted by P_t , and inflation is defined as $\Pi_t = P_t/P_{t-1}$ with steady state $\bar{\Pi}$.

With all these ingredients, the maximization problem of household h is stated as:

$$\max_{\{C_{ht}, D_{ht}, K_{ht}^H, H_{ht}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{(C_{ht} - bC_{ht-1})^{1-\sigma}}{1-\sigma} - \frac{\xi_H H_{ht}^{1+\varphi_H}}{1+\varphi_H} \right], \quad (1)$$

⁹See, for example, [Smets and Wouters \(2007\)](#), [Christiano et al. \(2005\)](#), and [Christiano, Motto, and Rostagno \(2014\)](#).

¹⁰Capturing the non-bank-dependent part of the economy prevents our model from overstating the macroeconomic consequences of changes in credit supply.

subject to the budget constraint:

$$\begin{aligned}
P_t C_{ht} + P_t D_{ht} + P_t(Q_t + \varsigma_t)K_{ht}^H &= W_{ht}H_{ht} \\
+ R_{Dt-1}P_{t-1}D_{ht-1} + P_t[z_t + (1 - \delta)Q_t]K_{ht-1}^H &+ \Sigma_{ht},
\end{aligned} \tag{2}$$

where Σ_{ht} summarizes other cash flows that the household receives, but which are irrelevant for its optimization problem. We assume that the household invests its deposits symmetrically in all the (symmetric) banks in the economy. Appendix A.1 provides the FOCs for this problem.

2.1.1 Nominal Wage Setting

The model features sticky wages.¹¹ A labor union collects all household-differentiated varieties of labor H_{ht} , which are sold to a competitive labor packer after setting nominal wages W_{ht} . The elasticity of substitution between varieties is ϵ_W . Wage setting is subject to Rotemberg (1982) adjustment costs governed by parameter θ_W which the union finances by charging households a lump-sum fee. Since these elements are standard in New Keynesian models, further details are relegated to Appendix A.2.

2.1.2 Bankers & Entrepreneurs

Bankers and entrepreneurs are modeled in a symmetric manner, and are therefore discussed together in this section.

At date t , bankers and entrepreneurs invest symmetrically in an all-islands portfolio of one-period banks and entrepreneurial firms, respectively. Bankers and entrepreneurs receive the terminal net worth of their banks and firms at the beginning of $t + 1$. At that point, bankers are also charged a lump sum tax by the government to finance the deposit insurance agency (DIA).

To make bankers and entrepreneurs net worth scarce, we assume that in every period a fraction $(1 - \theta_\chi)$, $\chi \in \{B, E\}$ of bankers and entrepreneurs retire and become workers, while the same measure of workers becomes bankers and entrepreneurs. When they retire, they

¹¹As discussed in, e.g., Galí (2015) and Smets and Wouters (2007), sticky wages contribute to dampen the rise in inflation after, e.g., an expansionary monetary policy shock, consistent with data. In our setup this adds realism to the impact of debt deflation on financial sector defaults.

pay out their wealth to households. New bankers and entrepreneurs in period t on the other hand receive a fraction ξ_χ of the net worth of the bankers and entrepreneurs that have retired in period t . Calibration of the parameters will ensure that equity is scarce enough for banks and entrepreneurs never to finance all their investments without debt.

In every period, continuing and new bankers and entrepreneurs decide how much (real) dividends ν_t^χ to pay out to their households and how much (real) equity χ_t to invest in the equity portfolio (where $\chi_t = B_t$ for bankers and $\chi_t = E_t$ for entrepreneurs, respectively). Bankers and entrepreneurs take the nominal return on their equity ρ_t^χ as given. Stating the maximization problem in real terms, the value function of a representative banker or entrepreneur i is

$$V_t^\chi(\chi_{it}) = \max_{\nu_{it}^\chi \geq 0, \chi_{it} \geq 0} \mathbb{E}_t \left[\nu_{it}^\chi + \mathbb{E}_t \Lambda_{t,t+1} \left((1 - \theta_\chi) \chi_{t+1} + \theta_\chi V_{t+1}^\chi(\chi_{it+1}) \right) \right], \quad (3)$$

where $\Lambda_{t,t+1}$ is the household's real stochastic discount factor, and

$$\chi_{it+1} = \frac{\rho_{t+1}^\chi}{\Pi_{t+1}} (\chi_{it} - \nu_{it}^\chi). \quad (4)$$

Following the established approach in the literature, we guess and verify that the value function is linear in the net worth of banker or entrepreneur i : $V_t^\chi(\chi_{it}) = s_t^\chi \chi_{it}$.¹² We further guess (and later verify) that in the vicinity of the steady state we have $s_t^\chi > 1$, which implies $\nu_t^\chi = 0$ by the Envelope Theorem.¹³ It then follows that

$$s_t^\chi = \mathbb{E}_t \underbrace{\Lambda_{t,t+1} (1 - \theta_\chi + \theta_\chi s_{t+1}^\chi)}_{=\Lambda_{t,t+1}^\chi} \frac{\rho_{t+1}^\chi}{\Pi_{t+1}}. \quad (5)$$

Equation (5) defines the bankers', or entrepreneurs', stochastic discount factor for later use as $\Lambda_{t,t+1}^\chi = \Lambda_{t,t+1} (1 - \theta_\chi + \theta_\chi s_{t+1}^\chi)$. Finally, the aggregate law of motion of equity of bankers or entrepreneurs is

$$\chi_{t+1} = [\theta_\chi + \xi_\chi (1 - \theta_\chi)] \frac{\rho_{t+1}^\chi}{\Pi_{t+1}} \chi_t - \frac{T_{t+1}}{P_{t+1}}, \quad (6)$$

¹²See, for example, [Gertler and Kiyotaki \(2010\)](#).

¹³We make sure that under our calibration of the model parameters $s_t^\chi = 1$ with a probability close to 0 and thus directly impose $\nu_t^\chi = 0$.

where T_{t+1} are nominal lump-sum taxes imposed by the deposit insurance agency, described below in detail.

2.2 Entrepreneurial Firms

Entrepreneurial firms provide the key connection between the financial sector and the real economy: they rely on bank loans to invest in physical capital used in the production sector. They hence transmit conditions in the financial sector to the real economy through their (physical) capital supply, and in turn transmit conditions in the real economy to the financial sector through the impact of the real return on (physical) capital on the loan default probability.

Each island is populated by a unit continuum of entrepreneurial firms indexed by j . These are one-period institutions owned by entrepreneurs. Entrepreneurial firms purchase physical capital K_{jt}^E from capital producers at real price Q_t . To finance their investment, they use loans L_{jt} from the bank on their island and entrepreneurial equity E_{jt} :

$$E_{jt} + L_{jt} = Q_t K_{jt}^E. \quad (7)$$

At date $t + 1$, entrepreneurial firms rent capital acquired at the end of t to intermediate good producers against a rental price z_{t+1} , and sell undepreciated capital $(1 - \delta)K_{jt}$ back to capital producers at real price Q_{t+1} .¹⁴ Following [Mendicino et al. \(2026\)](#), the final asset value of every entrepreneurial firm is subject to an idiosyncratic shock ω_j and an island specific shock ω_k , engendering loan default risk which is only partly diversifiable at the island-specific banks. By limited liability, the nominal terminal net worth of entrepreneurial firm j on island k at time $t + 1$ is

$$P_{t+1} \Omega_{jkt+1}^{Firm}(\omega_j, \omega_k) = \max\{\omega_j \omega_k [P_{t+1} Q_{t+1} (1 - \delta) K_{jt}^E + P_{t+1} z_{t+1} K_{jt}^E] - R_{Ljt} P_t L_{jt}, 0\}. \quad (8)$$

According to Equation (8), entrepreneurial firm j defaults at $t + 1$ if its idiosyncratic shock

¹⁴In contrast to models in which all production factors are pre-financed with loans (e.g., [Mendicino et al., 2026](#) and [Hristov and Hülsewig, 2017](#)), our setup with only pre-financed capital allows output at t to respond to the monetary policy rate, as in the canonical New Keynesian model.

is below the threshold $\bar{\omega}_{Ft+1}(\omega_k)$:

$$\bar{\omega}_{Ft+1}(\omega_k) = \frac{R_{Ljt}L_{jt}}{\omega_k \Pi_{t+1} [Q_{t+1}(1-\delta)K_{jt}^E + z_{t+1}K_{jt}^E]}. \quad (9)$$

To capture the impact of uncertainty on the fluctuation of default risk, we introduce [Christiano et al. \(2014\)](#) risk shocks in the same way as [Mendicino et al. \(2026\)](#). Specifically, we assume the shocks ω_j and ω_k are independent and log-normal distributed, with time-varying mean and variance:

$$\ln(\omega_{\Xi}) \sim N\left(-\frac{\sigma_{\omega_{\Xi t}}^2}{2}, \sigma_{\omega_{\Xi t}}^2\right), \quad \Xi \in \{j, k\}, \quad (10)$$

where the standard deviation $\sigma_{\omega_{\Xi t}}$ follows:

$$\ln(\sigma_{\omega_{\Xi t}}) = (1 - \rho_{\omega_{\Xi}})\ln(\bar{\sigma}_{\omega_{\Xi}}) + \rho_{\omega_{\Xi}}\ln(\sigma_{\omega_{\Xi t-1}}) + \sigma_{\sigma_{\Xi}}\epsilon_{\sigma_{\Xi}}, \quad \epsilon_{\sigma_{\Xi}} \sim N(0, 1). \quad (11)$$

While the risk shocks $\epsilon_{\sigma_{\Xi}}$ are mean preserving ($\mathbb{E}(\omega_{\Xi t}) = 1 \forall t$), a higher value of $\sigma_{\omega_{\Xi t}}^2$ implies that the distribution of ω 's has fatter tails, leading to higher default risk.

At the end of each period, all terminal net worth of entrepreneurial firms is paid out to entrepreneurs. The nominal return on entrepreneurial equity is:

$$\rho_{t+1}^E = \frac{\Pi_{t+1} \int_0^\infty \int_0^\infty \Omega_{jkt+1}^{Firm}(\omega_j, \omega_k) dF_{jt+1}(\omega_j) dF_{kt+1}(\omega_k)}{E_t}. \quad (12)$$

2.3 Banks

Each island is populated by a representative bank k . Like entrepreneurial firms, banks are active between two consecutive periods t and $t+1$. In period t banks combine equity B_{kt} from bankers and insured deposits D_{kt} from households in order to extend loans L_{kt} to entrepreneurial firms operating in their island. The bank can also invest in both one-period government bonds S_{kt} , remunerated at the policy rate R_t set by the central bank, and long-term zero-coupon bonds S_{kt}^L , trading at market price Q_t^S . The latter are purchased by banks from a *bond management company* in period t and, if not maturing, resold to it in period $t+1$. The role of this company is discussed below in detail. For tractability, as in [Hatchondo and Martinez \(2009\)](#) and [Chatterjee and Eyigungor \(2012\)](#), these bonds are assumed to reach

maturity in an independent random manner with probability $\frac{1}{m}$ per period so that in each period a fraction $\frac{1}{m}$ of them mature. This implies that the average maturity of the bonds is m periods. In market value terms, banks face the following balance-sheet constraint

$$L_{kt} + S_{kt} + Q_t^S S_{kt}^L = B_{kt} + D_{kt}. \quad (13)$$

At the beginning of period t , every bank k is endowed with an identical amount of equity $\bar{B}_t$ by bankers, such that $B_{kt} = \bar{B}_t$.

Banks extend loans in a perfectly competitive manner. It is assumed that in addition to the loan rate R_{Lt} , banks reap a non-pecuniary benefit c_R per unit of lending, reflecting the value of lending relationships (recent empirical evidence of the value of such relationships to banks is provided in [Agarwal, Chomsisengphet, Liu, Song, and Souleles \(2018\)](#), with [Boot \(2000\)](#) surveying the theoretical underpinnings). It is thus costly for the bank to reduce lending beyond the bank's lending capacity implied by capital requirements (to be discussed below), reflecting a weakening in lending relationships.¹⁵ The role of this assumption will be discussed below, alongside the bank's profit maximization problem.

Banks raise insured deposits D_{kt} in a monopolistic manner at rate R_{Dkt} . The demand for deposits of bank k is:

$$D(R_{Dkt}, R_{Dt}) = \left(\frac{R_{Dkt}}{R_{Dt}} \right)^{-\epsilon_D} D_t,$$

$$\epsilon_D < -1. \quad (14)$$

Banks take the aggregate deposit rate

$$R_{Dt} = \left(\int_0^1 R_{Dkt}^{1-\epsilon_D} dk \right)^{\frac{1}{1-\epsilon_D}}$$

as given. This structure implies that deposit rate setting exhibits imperfect competition and

¹⁵While each bank is a one-period institution, the bank's shareholders continue to profit from the banking relationship in the future. While we do not model such dynamics, this can for example be thought of as relationships embodied in the employees of the bank k (documented in [D'Andrea and Stivella, 2025](#)), not in the institution per se, with a constant set of employees employed by the succession of banks on island k . Each bank is managed in the interest of shareholders, and internalizes the benefits of the lending relationship in this manner, but cannot use it to avoid default due to its non-pecuniary nature.

delivers a sensitivity of deposit rates to the policy rate R_t consistent with [Drechsler et al. \(2021\)](#), namely that deposit spreads are increasing in the policy rate.

In line with the deposit franchise framework of [Drechsler et al. \(2021\)](#), banks incur a per-period operating cost. As discussed therein, the operating cost has two important implications. First, it allows to view the deposit franchise as an implicit interest rate swap. Second, its presence helps the model generate empirically plausible levels of return on bank equity.

We model this operating cost as $C_{ft} = c_f L_t$ (in real terms). While the deposit franchise is conceptually tied to deposits—since deposit funding generates a margin $(R_t - R_{Dkt})D_{kt}$ —in general equilibrium deposits are tightly linked to lending through the balance sheet identity, implying that D_t is proportional to L_t , up to fluctuations in leverage. Motivated by this relationship, we express the franchise in units of lending, which provides a parsimonious normalization in a setting where bank size is naturally measured by total lending.

Under this representation, the net payoff from deposit funding can be written as

$$L_t \left[(R_t - R_{Dkt}) \frac{D_t}{L_t} - c_f \right]$$

highlighting that the relevant object is the deposit spread, expressed per unit of lending. This formulation preserves the economic role of the fixed cost emphasized by [Drechsler et al. \(2021\)](#), while maintaining model stability. This is because a strictly fixed operating cost would not adjust with bank size and can lead to dynamics in which banks are unable to rebuild equity when capital becomes scarce, thereby generating instability and counterfactual corner solutions with zero aggregate lending. By contrast, the proportional specification preserves increasing returns to equity in such states and ensures well-behaved equilibrium dynamics, including satisfaction of the Blanchard–Kahn conditions.

Finally, to allow the model to jointly match (i) the average spread between deposit rates and the policy rate and (ii) the low pass-through of policy rates to deposit rates documented by [Drechsler et al. \(2021\)](#), we introduce adjustment costs in deposit rate setting in the spirit of [Gerali, Neri, Sessa, and Signoretti \(2010\)](#). Each bank k faces a quadratic cost

$$\frac{\kappa_D}{2} \left(\frac{R_{Dkt}}{R_{Dkt-1}} - 1 \right)^2 R_{Dkt} D_{kt}$$

which penalizes deviations from the deposit rate of the previous period. This friction slows down the adjustment of deposit rates and is key to reproducing the empirically observed sluggishness in deposit pass-through. A more detailed discussion is provided when deriving the optimal deposit rate condition.

Intertemporal trade of the long-term bonds. The banking industry operates a long-term bond management company which centralizes the trade of bonds between the subsequent cohort of banks. The main role of this company is to keep track of the book value of the bonds to replicate the situation in which bonds were held by banks operating over multiple periods. At date t , this company buys the bonds at (real) market price Q_t^S from surviving banks that bought them at $t - 1$ and from the Deposit Insurance Agency (DIA), which repossesses the bonds from failing banks that bought them at $t - 1$. Then, the company sells the bonds at market price to the new cohort of banks that buy them (together with the newly issued long-term bonds) at t .

Importantly, this company provides a “certificate of book value” to the bonds, which allows banks to write their balance sheet for regulatory purposes (in real terms) as follows:

$$L_{kt} + S_{kt} + Q_t^{BV} S_{kt}^L = \bar{B}_t + D_{kt} + (Q_t^{BV} - Q_t^S) S_{kt}^L, \quad (15)$$

where Q_t^{BV} is the real average book value of bonds according to the certificate, and $(Q_t^{BV} - Q_t^S) S_{kt}^L$ measures what would be regarded as unrealized capital losses (if $Q_t^{BV} > Q_t^S$) or gains (if $Q_t^{BV} < Q_t^S$) if the bank were measuring the value of its bonds at their certified book value. Given the permanent inventory dynamics of the stock of long-term bonds, the law of motion of Q_t^{BV} is given by:

$$Q_t^{BV} S_{kt}^L = \frac{Q_{t-1}^{BV}}{\Pi_t} \left(1 - \frac{1}{m}\right) S_{kt-1}^L + Q_t^S \left[S_{kt}^L - \left(1 - \frac{1}{m}\right) S_{kt-1}^L \right], \quad (16)$$

where the first term in the right hand side represents the continuation book value of the non-matured bonds, and the second term is the market value of the newly issued bonds acquired by the bond management company in the primary bond market.¹⁶

¹⁶Note that the book value of the newly issued bonds coincides with the market value.

Capital requirement. Banks are subject to a minimum capital requirement, which imposes that banks must operate with *regulatory* equity capital greater than or equal to a fraction γ of their loans. The key novelty in our analysis is the comparison of capital requirements under two different definitions of regulatory capital. Under a fair-value definition, the capital requirement is of the following form:

$$\gamma L_{kt} \leq \bar{B}_t. \quad (17)$$

Under the book-value definition, the requirement takes the following form:

$$\gamma L_{kt} \leq B_{kt}^{BV}, \quad (18)$$

where

$$B_{kt}^{BV} \equiv \bar{B}_t + (Q_t^{BV} - Q_t^S)S_{kt}^L \quad (19)$$

represents the book value of bank equity, which is defined using Equation (15). Therefore, the difference between the two resulting capital requirements arises from the prudential treatment of *unrealized* gains and losses associated with banks' long-term bonds portfolio.

Terminal net worth of a bank. As derived in Equation (9), conditional on the island-idiosyncratic shock ω_k , an entrepreneurial firm pays back its loan in full when it experiences a firm-idiosyncratic shock no lower than $\bar{\omega}_{Ft+1}(\omega_k)$. In case of default of an entrepreneurial firm, the bank only recovers a fraction $(1 - \delta_M)$ of the firm's terminal asset value in Equation (8), where δ_M is an asset repossession cost. Hence, the nominal ex-post gross rate of return on loans of the bank in island k is

$$\begin{aligned} \tilde{R}_{Lkt+1}(\omega_k) = & \frac{\omega_k(1 - \delta_M)\Pi_{t+1}[Q_{t+1}(1 - \delta) + z_{t+1}]K_t^E}{L_t} \int_0^{\bar{\omega}_{Ft+1}(\omega_k)} \omega_j dF_{jt+1}(\omega_j) \\ & + R_{Lkt} \int_{\bar{\omega}_{Ft+1}(\omega_k)}^{\infty} dF_{jt+1}(\omega_j), \end{aligned} \quad (20)$$

where K_t^E denotes the aggregate level of physical capital held by entrepreneurs. By definition of the entrepreneurial firm's default threshold in Equation (9), the first term is bounded by R_{Lkt} : when borrowers default, they repay less than the agreed loan rate, otherwise they repay

fully. Thus the model features a structural link between bank asset returns and borrower defaults, which is crucial to accurately capture the importance of the feedback loop between conditions in the real economy and the financial sector (see [Mendicino et al., 2026](#)).

Due to the stochastic maturity of the long-term bond portfolio, the nominal gross rate of this portfolio is:

$$R_{kt+1}^S = \frac{\frac{1}{m} + (1 - \frac{1}{m}) \Pi_{t+1} Q_{t+1}^S}{Q_t^S}. \quad (21)$$

The nominal terminal net worth of the bank on island k is then

$$\begin{aligned} P_{t+1} \Omega_{kt+1}^B(\omega_k) = P_t \left[\tilde{R}_{Lkt+1}(\omega_k) L_{kt} + R_t S_{kt} + R_{kt+1}^S Q_t^S S_{kt}^L - C_{ft} \right. \\ \left. - R_{Dkt} D_{kt} - \frac{\kappa_D}{2} \left(\frac{R_{Dkt}}{R_{Dkt-1}} - 1 \right)^2 R_{Dkt} D_{kt} \right]. \end{aligned} \quad (22)$$

Banks default on their deposits if their terminal net worth is negative. From Equation (22), it is useful to define a threshold value for the island-specific shock ω_k below which the bank in island k defaults. This is implicitly done in the next equation

$$\tilde{R}_{Lkt+1}(\bar{\omega}_{bkt+1}) L_{kt} + R_t S_{kt} + R_{kt+1}^S Q_t^S S_{kt}^L - R_{Dkt} D_{kt} - C_{ft} - \frac{\kappa_D}{2} \left(\frac{R_{Dkt}}{R_{Dkt-1}} - 1 \right)^2 R_{Dkt} D_{kt} = 0. \quad (23)$$

Equation (23) implies that the banks' failure rate at the beginning of period $t + 1$ is $F_{kt+1}(\bar{\omega}_{bkt+1})$. Thus, the nominal gross rate of return on the portfolio of equity of a banker that symmetrically invests in all banks is

$$\rho_{t+1}^B = \frac{\Pi_{t+1} \int_{\bar{\omega}_{bkt+1}}^{\infty} \Omega_{kt+1}^B(\omega_k) dF_{kt+1}(\omega_k)}{\bar{B}_t}. \quad (24)$$

Bank's profit maximization problem. Banks are managed in the interest of bankers, and maximize:

$$\begin{aligned} \max_{L_{kt}, S_{kt}, S_{kt}^L, R_{Dkt}} \mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \left\{ c_R L_{kt} + \int_0^{\infty} \max \left[\tilde{R}_{Lkt+1}(\omega) L_{kt} + R_t S_{kt} + \right. \right. \\ \left. \left. R_{kt+1}^S Q_t^S S_{kt}^L - C_{ft} - \left[1 + \frac{\kappa_D}{2} \left(\frac{R_{Dkt}}{R_{Dkt-1}} - 1 \right)^2 \right] R_{Dkt} D_{kt}, 0 \right] F_{kt+1}(\omega) \right\} \end{aligned} \quad (25)$$

subject to

$$L_{kt} + S_{kt} + Q_t^S S_{kt}^L = D_{kt} + \bar{B}_t, \quad (26)$$

$$D_{kt} = \left(\frac{R_{Dkt}}{R_{Dt}} \right)^{-\epsilon_D} D_t, \quad (27)$$

and under fair-value capital requirements:

$$\gamma L_{kt} \leq \bar{B}_t, \quad (28)$$

while under book-value requirements:

$$\gamma L_{kt} \leq B_{kt}^{BV}. \quad (29)$$

First, the problem implies the following arbitrage condition for the two bond types:

$$\mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} [R_{kt+1}^S - R_t] [1 - F_{kt+1}(\bar{\omega}_{bkt+1})] = 0. \quad (30)$$

The expression shows the close link between the policy rate and the yield of the long-term bond portfolio. This relationship is central to the effects of monetary policy on unrealized gains or losses that banks' faces. In line with the empirical evidence in [Marsh and Laliberte \(2023\)](#), it implies that banks in the model face valuation losses on their long-term bond portfolio when the policy rate rises.

Next, the optimal deposit rate R_{Dt} is independent of all other choices, and is determined by:

$$-1 + \epsilon_D - \epsilon_D(R_t - R_{Dkt}) - \kappa_D \left(\frac{R_{Dkt}}{R_{Dkt-1}} - 1 \right) \frac{R_{Dkt}}{R_{Dkt-1}} = 0 \quad (31)$$

We restrict attention to a symmetric equilibrium in which all banks offer the same deposit rate in every period. This implies $R_{Dt} = R_{Dkt}$, as well as $D_t = D_{kt}$.

To understand the role of adjustment costs, note that in their absence, the deposit rate would be given by

$$R_{Dt}^{noAdj} = \frac{\epsilon_D}{\epsilon_D - 1} R_t$$

In order for the model to reproduce realistic deposit spreads, the ratio $\frac{\epsilon_D}{\epsilon_D - 1} < 1$ has to be

close to 1. However, since in the case without adjustment costs the ratio coincides with the sensitivity of the deposit rate to the policy rate (i.e., the deposit beta), this is at odds with empirical evidence of a deposit beta well below 1 (Drechsler et al., 2021; Bonfim and Queiró, 2024). Adjustment costs for deposit rates allow the model to reproduce simultaneously a realistic deposit spread and a realistic sensitivity of the deposit rate to the policy rate. This is important in our context, since Drechsler et al. (2021) argue that the relative insensitivity of deposit rates to changes in the policy rate hedges interest rate risk exposure of banks.

Furthermore, it can be proven that for any deposit rate (including the optimal deposit rate), the objective function is convex in the loan volume L_{kt} (but not necessarily strictly convex). Banks are indifferent between any $L_{kt} \in [0, \bar{L}_t]$ (where $\bar{L}_t = \frac{\bar{B}_t}{\gamma}$ under fair-value capital requirements or $\bar{L}_t = \frac{B_t^{BV}}{\gamma}$ under book-value capital requirements) if two conditions are simultaneously satisfied: if (i) a bank makes profits on non-lending activities that are so high that the bank never fails for any feasible loan volume $L < \bar{L}_t$, and (ii) the loan rate fulfills:

$$\mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \left[c_R + \int_{\bar{\omega}_{bkt+1}}^{\infty} [\tilde{R}_{L_{t+1}}(\omega) - R_t] dF_{kt+1}(\omega) \right] = 0. \quad (32)$$

Otherwise, there is a corner solution and either $L_{kt}^* = 0$ or the bank chooses the maximum loan volume it can extend without violating the capital requirement, i.e. $L_{kt}^* = \frac{\bar{B}_t}{\gamma}$ under fair-value based capital requirements, and $L_{kt}^* = \frac{B_t^{BV}}{\gamma}$ under book-value based requirements. Which corner is optimal for the bank depends on the loan rate $R_{L_{kt}}$, which the bank takes as given. The proof can be found in the Appendix. In summary, banks strictly prefer extending loans if the expected ex-post return is such that:

$$\begin{aligned} & \mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \int_{\bar{\omega}_{bkt+1}}^{\infty} \left\{ [\tilde{R}_{L_{t+1}}(\omega) - R_t(1 - \gamma)] \bar{L}_t + (R_t - R_{Dt}) D_t - \frac{\kappa_D}{2} \left(\frac{R_{Dt}}{R_{Dt-1}} - 1 \right)^2 R_{Dt} D_t \right. \\ & \quad \left. + (R_{t+1}^S - R_t) Q_t^S S_{kt}^L - C_{ft} \right\} dF_{kt+1}(\omega) \\ & > \mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \left[(R_{t+1}^S - R_t) Q_t^S S_{kt}^L + (R_t - R_{Dt}) D_t - \frac{\kappa_D}{2} \left(\frac{R_{Dt}}{R_{Dt-1}} - 1 \right)^2 R_{Dt} D_t - C_{ft} + R_t \bar{B}_t - c_R \bar{L}_t \right]. \end{aligned} \quad (33)$$

This condition is verified numerically under the calibration explained in the next section.

The role of the non-pecuniary benefit c_R is to make sure that under that calibration banks don't switch between extending zero loans and operating at maximum lending capacity, a pattern not observed in the data.¹⁷

Assuming that (33) is fulfilled, all banks choose identical loan volumes $\bar{L}_t$, as they receive the same level of equity from bankers. The k index is therefore dropped in continuation. Bank payoffs at time t are then a function of the maximum loan volume $\bar{L}_{t-1}$ the bank can extend at time $t - 1$ to comply with capital requirements, and are given by:

$$\begin{aligned} \Omega_t^B(\bar{L}_{t-1}) = & \left[(R_{t-1} - R_{Dt-1})D_{t-1} + \frac{\kappa_D}{2} \left(\frac{R_{Dt-1}}{R_{Dt-2}} - 1 \right)^2 R_{Dt-1}D_{t-1} \right. \\ & \left. - C_{ft-1} + (R_t^S - R_{t-1})Q_{t-1}^S S_{t-1}^L \right] (1 - F_{kt}(\bar{\omega}_{kt})) + \int_{\bar{\omega}_{bt}}^{\infty} \left[(\tilde{R}_{Lkt}(\omega) - R_{t-1}(1 - \gamma))\bar{L}_{t-1} \right] dF_{kt}(\omega). \end{aligned} \quad (34)$$

Deposit insurance agency. The DIA supervises the liquidation process of failed-bank assets, which is subject to proportional repossession costs δ_B .¹⁸ It imposes a nominal lump-sum tax T_{t+1} on bankers to (ex-post) balance its budget period-by-period. The total nominal lump sum tax T_{t+1} is

$$\begin{aligned} \frac{T_{t+1}}{P_t} = & \left[R_{Dt}D_t + \frac{\kappa_D}{2} \left(\frac{R_{Dt}}{R_{Dt-1}} - 1 \right)^2 R_{Dt}D_t + C_{ft} - R_{t+1}^S Q_t^S S_t^L - R_t S_t \right] F_{kt+1}(\bar{\omega}_{kt+1}) \\ & - (1 - \delta_B) \left[\int_0^{\bar{\omega}_{bt+1}} \tilde{R}_{Lt+1}(\omega) L_t dF_{kt+1}(\omega) \right]. \end{aligned} \quad (35)$$

2.4 Contracting Problem Between Firms & Banks

Entrepreneurial firms enter a contract with the bank on their island k that specifies the loan rate and the leverage of entrepreneurs (or equivalently: the loan rate, the loan volume and the total amount of capital bought). Bankers are indifferent between any combination of

¹⁷If c_R were set to zero, equity would need to be made artificially scarce for (33) to hold in every period, implying counterfactually large loan spreads.

¹⁸The model follows [Bernanke, Gertler, and Gilchrist \(1999\)](#) in adopting a ‘‘costly state verification’’ setup, by which the DIA must incur a cost that is proportional to the assets of the bank in order to observe the realization of the idiosyncratic shocks.

loan rates and leverage on their iso-expected-profit curve:

$$\begin{aligned} \bar{\Omega}_t^b = \mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \int_{\bar{\omega}_{bt+1}}^{\infty} & \left[(\tilde{R}_{Lt+1}(\omega) - R_t(1 - \gamma))L_t + (R_t - R_{Dt})D_t \right. \\ & \left. - \frac{\kappa_D}{2} \left(\frac{R_{Dt}}{R_{Dt-1}} - 1 \right)^2 R_{Dt}D_t + (R_{t+1}^S - R_t)Q_t^S S_{kt}^L - C_{ft} \right] dF_{kt+1}(\omega). \end{aligned} \quad (36)$$

Entrepreneurial firm jk active from time t to $t+1$ maximizes its properly discounted value for entrepreneurs $\mathbb{E}_t(\Lambda_{t,t+1}^E \Omega_{jkt+1}^{Firm})$, by choosing a point on the bank's isoprofit curve.¹⁹ From the perspective of the firms, the total loan volume L_t intermediated by each bank is exogenous (and given by either $\frac{B_t}{\gamma}$ or $\frac{B_t^{BV}}{\gamma}$ depending on the type of capital requirements). As in [Mendicino et al. \(2026\)](#), firms also take the bank's default cutoff as given. Using Eq. (5), in equilibrium it must be that $s_t \bar{B}_t = \bar{\Omega}_t^b$. The contracting problem between the entrepreneurial firm and the bank on island k is then given by:

$$\begin{aligned} \max_{K_{jkt}, L_{jkt}, R_{Ljkt}} & \mathbb{E}_t \Lambda_{t,t+1}^E \Omega_{jkt+1}^{Firm} \\ \text{subject to} & \text{ Eq. (7), Eq. (36).} \end{aligned} \quad (37)$$

The FOCs are presented in [Appendix A.7](#).

2.5 Fiscal Policy

As discussed above, there are two types of government bonds in this model: one-period bonds and long-term bonds. Both are assumed to be in fixed real supply: for the one-period bonds we assume a zero net-supply, while for the long-term bonds we assume a positive real supply S^L , a parameter to be calibrated. To introduce a demand side shock, we also assume that the fiscal authority engages in government spending G_t , governed by the following AR(1) process:

$$\ln(G_t) = (1 - \rho_g)\ln(G) + \rho_g \ln(G_{t-1}) + \sigma_g \epsilon_{Gt}, \quad \epsilon_{Gt} \sim N(0, 1). \quad (38)$$

¹⁹There is no reason for entrepreneurial firms to make bankers better off than necessary for them to participate, therefore the contract they offer lies on the bank's iso-expected-profit curve.

We assume that the government balances its budget in every period by charging households a lump-sum tax.

2.6 Monetary Policy

We assume that there is a central bank which sets the nominal gross policy rate R_t according to the following Taylor rule:

$$R_t = R^{1-\phi_R} R_{t-1}^{\phi_R} \left(\frac{\Pi_t}{\bar{\Pi}} \right)^{\phi_\pi(1-\phi_R)} \left(\frac{Y_t}{Y_{t-1}} \right)^{\phi_Y(1-\phi_R)} \tau_t, \quad (39)$$

where R is the long-term target monetary policy rate, and $\bar{\Pi}$ is steady state inflation. ϕ_R is a smoothing parameter, while ϕ_Y and ϕ_π govern how strongly the central bank reacts to deviations from GDP and inflation, respectively. τ_t is a monetary policy shock evolving according to

$$\ln(\tau_t) = \rho_\tau \ln(\tau_{t-1}) + \sigma_\tau \epsilon_{\tau t}, \quad \epsilon_{\tau t} \sim N(0, 1). \quad (40)$$

2.7 Production

The description of the production side of the economy follows a standard New Keynesian formulation and its full description is relegated to Appendix A.3. Here, it shall suffice to state a few elements. The aggregate production function is

$$Y_t = \theta_t K_t^\alpha H_t^{1-\alpha}, \text{ with } \alpha \in [0, 1], \quad (41)$$

where K_t and H_t are aggregate physical capital and aggregate labor hours, respectively, and aggregate productivity θ_t is stochastic and follows an AR(1) process:

$$\ln(\theta_t) = \rho_\theta \ln(\theta_{t-1}) + \sigma_\theta \epsilon_{\theta t}, \quad \epsilon_{\theta t} \sim N(0, 1). \quad (42)$$

The New Keynesian Phillips Curve of the model, arising from the problem of a unit continuum of final good producers facing a stochastic elasticity of substitution between

intermediate goods μ_t , as well as [Rotemberg \(1982\)](#) price adjustment costs is

$$\theta_R \left(\frac{\Pi_t}{\bar{\Pi}} - 1 \right) \frac{\Pi_t}{\bar{\Pi}} = \mathbb{E}_t \Lambda_{t,t+1} \theta_R \left(\frac{\Pi_{t+1}}{\bar{\Pi}} - 1 \right) \frac{\Pi_{t+1}}{\bar{\Pi}} \frac{Y_{t+1}}{Y_t} + mc_t \mu_t + (1 - \mu_t). \quad (43)$$

The stochastic elasticity of substitution μ_t also follows an AR(1) process:

$$\ln(\mu_t) = (1 - \rho_\mu) \ln(\mu) + \rho_\mu \ln(\mu_{t-1}) + \sigma_\mu \epsilon_{\mu t}, \quad \epsilon_{\mu t} \sim N(0, 1). \quad (44)$$

Physical capital is produced by combining the final good with undepreciated capital, subject to an adjustment cost of $\mathcal{C} \left(\frac{I_t}{K_{t-1}} \right)$ as in [Jermann \(1998\)](#).²⁰ The aggregate capital stock evolves according to:

$$K_t = I_t + (1 - \delta) K_{t-1} + \mathcal{C} \left(\frac{I_t}{K_{t-1}} \right) K_{t-1}. \quad (45)$$

2.8 Capital Management Firms

There is a unit continuum of competitive capital management firms. They charge households a fee ς_t per unit of capital, and face costs of $\frac{\kappa_H}{2} (K_t^H)^2$. Their maximization problem is:

$$\max_{K_t^H} \varsigma_t K_t^H - \frac{\kappa_H}{2} (K_t^H)^2. \quad (46)$$

3 Solution, Estimation, and Model Validation

This section outlines the computational method used to obtain the numerical solution of the model, discusses the calibration strategy, and explores the quantitative properties of the model.

3.1 Solution Method

The model is solved around the zero inflation steady state ($\bar{\Pi} = 1$), in line with much of the New Keynesian literature (see [Galí, 2015](#)). We employ third-order perturbation methods to

²⁰The functional form is $\mathcal{C} \left(\frac{I_t}{K_{t-1}} \right) = \frac{a_{k,1}}{1 - \frac{1}{\kappa}} \left(\frac{I_t}{K_{t-1}} \right)^{1 - \frac{1}{\kappa}} + a_{k,2}$. While κ is estimated, $a_{k,1}$ and $a_{k,2}$ are set such that in steady state $I = \delta K$ and $\mathcal{C}(\delta) = 1$.

obtain an approximation of the policy functions around the deterministic steady state.²¹ The integrals involving the realized ex-post returns on bank loans (as well their derivatives) cannot be written as an explicit function of the state variables, which introduces a complication. We follow [Mendicino et al. \(2026\)](#) in overcoming this challenge by approximating the integrals by a sum of third-order Taylor approximations. More details are provided in Appendix [A.7](#).

3.2 Model Estimation

The model is calibrated to quarterly euro-area data from 1995 Q1 to 2016 Q4. Following standard practices, the calibration of the model proceeds in two steps.²²

First step. The coefficient of relative risk aversion σ is set to 1, which implies log-utility, the Frisch elasticity of labor supply φ_H to 1, the capital-share parameter of the intermediate goods production function α to 0.25, and the value of capital depreciation δ to 0.025. The elasticity of substitution parameters for differentiated labor services ϵ_W and final goods μ are set to 5 and 7.25, respectively, resulting in a wage markup of 20% ([Smets and Wouters, 2003](#)), and a markup of 16% in the goods market which is consistent with euro-area estimates reported in [Christopoulou and Vermeulen \(2012\)](#). The scaling parameter ξ_N associated with labor disutility is set to normalize a steady-state labor supply of $H = 1$. Following [Stähler and Thomas \(2012\)](#), the steady state government spending G is set to 22.56% of GDP. Following [Born and Pfeifer \(2014\)](#), we estimate the government spending shocks externally by OLS on Equation (38) (in logs).

The maturity of long-term bonds m is set to 13.6, which implies an average maturity of bank bond holdings of 3.4 years ([Hoffmann, Langfield, Pierobon, and Vuillemeys, 2019](#)). The value of bankruptcy parameters δ_B and δ_M are both set equal to 0.30, in line with empirical studies (e.g. [Alderson and Betker, 1995](#); [Djankov, Hart, McLiesh, and Shleifer, 2008](#); [Granja, Matvos, and Seru, 2017](#)). We set both θ_B and θ_E to 0.975, implying that

²¹See [Christiano et al. \(2014\)](#) and [Born and Pfeifer \(2014\)](#) for discussions on the necessity of third-order approximations to appropriately capture the effects of volatility shocks, such as those affecting the cumulative distribution functions of the firm-idiosyncratic and island-specific shocks.

²²See e.g. [Fernández-Villaverde, Guerrón-Quintana, Rubio-Ramírez, and Uribe \(2011\)](#) and [Born and Pfeifer \(2014\)](#) for examples of DSGE models estimated in a two-step procedure using the Simulated Method of Moments.

bankers and entrepreneurs remain active for ten years on average. Finally, the minimum capital requirement γ is set to 0.08, consistent with the requirement under Basel II.

Second step. The remaining parameters are estimated using the Simulated Method of Moments (SMM).^{23,24} We obtain the model implied moments by simulating our baseline economy, under which banks' regulatory capital is defined on the basis of the book-value of their bonds portfolio. Our estimation targets include a number of macroeconomic, financial, and banking moments. We target the standard deviations and first two auto-correlations of GDP, consumption, investment, inflation, wages, the policy rate and labor hours, as well as their correlation with GDP. We also target a range of moments related to financial markets. These are the mean and standard deviation of the conditional expectation of firm and bank default rates and the unconditional correlation between the two default probabilities, the mean and standard deviation of the loan rate spread, the average deposit rate spread, the average central bank policy rate, the average aggregate loan to GDP ratio, the share of physical capital owned by households, and the average ratio of loans to bonds on bank's balance sheets. We also target the sensitivity of deposit rates to the policy rate, given its importance for the extent to which the deposit franchise hedges banks' interest rate risk (Drechsler et al., 2021). This sensitivity, commonly known as the deposit beta, is measured as the coefficient from a linear regression of the deposit rate on the policy rate. For Euro Area household deposits, Bonfim and Queiró (2024) report a deposit beta of 0.67. Finally, we target the response of aggregate economic activity to large declines in banking-sector equity. Baron, Verner, and Xiong (2021) find that a 30% decline in bank equity predicts, on average, a 4% decline in output within one year.

Tables 1 and 2 provide the values of moments targeted in the data, and compare them to their model generated counterparts. Parameter values are reported in the Appendix (Table 6). We obtain data on GDP, consumption, investment, government spending, total wages, hours worked, the GDP deflator and population from the OECD Quarterly National Accounts. Data on financial corporation loan volumes, loan rates and riskless interest rates are

²³The set of estimated parameters is $\beta, \xi_B, \xi_E, \sigma_{\omega_k}, \rho_{\omega_k}, \rho_{\omega_j}, \sigma_{\omega_j}, \bar{\sigma}_{\omega_k}, \bar{\sigma}_{\omega_j}, \sigma_\theta, \rho_\theta, \sigma_\tau, \rho_\tau, \sigma_\mu, \rho_\mu, b, \theta_R, \theta_W, \kappa, \kappa_H, \kappa_D, \phi_R, \phi_\Pi, \phi_Y, S^L$.

²⁴The good properties of SMM for estimation of non-linear DSGE models have been established in Ruge-Murcia (2012).

obtained from the ECB Statistical Data Warehouse.²⁵ All series are adjusted for seasonality

Table 1: *Calibration targets and model fit (macroeconomic)*

	$\sigma_{x_t}/\sigma_{\Delta y_t}$ (Y: $\sigma_{\Delta y_t}$)		$\rho(x_t, \Delta y_t)$	
	Data	Model	Data	Model
ΔY	1.186	1.246	1	1
ΔC	0.746	0.789	0.915	0.513
ΔI	2.594	2.833	0.93	0.638
Π	0.229	0.451	0.326	0.263
Δw	0.478	0.403	-0.164	-0.093
R	0.308	0.235	0.489	0.372
H	1.828	3.03	0.217	0.596
	$\rho(x_t, x_{t-1})$		$\rho(x_t, x_{t-2})$	
	Data	Model	Data	Model
ΔY	0.905	0.768	0.705	0.546
ΔC	0.888	0.918	0.702	0.747
ΔI	0.872	0.948	0.754	0.826
Π	0.437	0.798	0.372	0.671
Δw	0.868	0.831	0.787	0.585
R	0.971	0.912	0.922	0.774
H	0.915	0.952	0.837	0.899

Notes: All series are seasonally adjusted and all variables except Π and R are in logs. Δ^{HP} indicates the Hodrick-Prescott filter with HP parameter 1600. Data sources and variable definitions are described in Appendix B.

and those series that exhibit trends are detrended using the Hodrick-Prescott filter.²⁶ The moments for the mean and standard deviation of firm and bank defaults, the correlation between firm and bank defaults, and the mean and standard deviation of the rate of return on

²⁵The loan volume series include all available maturities. The loan spread is constructed as a volume weighted average over maturities, where the riskless rate is taken as the short term rate published by the ECB for maturities less than 1 year, the 2-year yield on triple A euro-area government bonds (published by the ECB) for maturities between 1 and 5 years, and the corresponding 5-year yield for maturities over 5 years.

²⁶Given that our data is quarterly, following Ravn and Uhlig (2002), we set the HP parameter to 1600. For those series that are not directly available with seasonal adjustment, such as loans, seasonal adjustment is done using X13-ARIMA using the R package seasonal.

Table 2: Calibration targets and model fit (financial)

Moment	Description	Data	Model
$100\mathbb{E}_t \left[\int_0^\infty F_{jt+1}(\bar{\omega}_{Ft+1}(\omega_k)) d\omega_k \right]$	Mean Firm Default	0.662	0.657
$100\mathbb{E}_{F_{kt+1}}(\bar{\omega}_{bt+1})$	Mean Bank Default	0.166	0.176
$\rho \left[\int_0^\infty F_{jt+1}(\bar{\omega}_{Ft+1}(\omega_k)) d\omega_k, F_{kt+1}(\bar{\omega}_{bt+1}) \right]$	Corr(Firm D., Bank D.)	0.642	0.642
$100\sigma \left[\int_0^\infty F_{jt+1}(\bar{\omega}_{Ft+1}(\omega_k)) d\omega_k \right]$	Std Firm Def.	0.549	0.287
$100\sigma [F_{kt+1}(\bar{\omega}_{bt+1})]$	Std Bank Def.	0.422	0.261
$100\sigma_{\rho_t^B}$	Std ROE Banks	2.06	2.76
$100(\mathbb{E}\rho_t^B - 1)$	Mean ROE Banks	1.6	2.52
$\mathbb{E}\frac{L_t}{Y_t}$	Mean L/Y	2.35	2.4
$100\mathbb{E}(R_{Lt} - R_t)$	Mean Loan spread	0.549	0.57
$\sigma_{(R_{Lt}-R_t)}/\sigma_{\Delta Y_t}$	Sd Loan spread	0.137	0.193
$100\mathbb{E}(R_t - 1)$	Mean Net Policy Rate	0.429	0.496
$\mathbb{E}(L_t/S_t^L)$	Loan-to-Bond Ratio	3.6	3.73
$\mathbb{E}(K_{Ht}/K_t)$	HH Capital Share	0.22	0.22
$100\mathbb{E}(R_t - R_{Dt})$	Deposit Spread	0.25	0.25
$\mathbb{E} \frac{\Delta_4 \log(Y_{t+4})}{\Delta_4 \log(EQ_t)} \mid p_1(\Delta_4 \log(EQ_t))$	GDP Loss From Large Equity Loss	-0.133	-0.144
β_D	Deposit beta	0.67	0.709

Notes: All series are seasonally adjusted. Interest rates, spreads, equity returns and default rates are reported at quarterly frequency. Δ^{HP} indicates the Hodrick-Prescott filter with HP parameter 1600. $\Delta_4 x_t$ indicates the fourth difference (that is, one-year difference) $x_t - x_{t-4}$, and $p_1(x)$ indicates that x is below its first percentile. Data sources and variable definitions are described in Appendix B.

bank equity are taken from Mendicino et al. (2026).²⁷ Appendix B contains further details on the data sources and construction, and the calibration strategy.

Overall, the model fits the data reasonably well. In particular, it closely matches banks' loans-to-bonds ratio of 3.6 and a deposit beta of 0.67, both of which are important for capturing banks' exposure to interest rate risk.

3.3 Model Validation

In this section, we assess the model's ability to reproduce empirical evidence on the loan-pricing effects of realized capital losses and unrealized balance-sheet losses. This exercise is useful because it evaluates the model along an important dimension for our counterfactual

²⁷The moments are based on euro area data from 1992Q1:2016Q4.

analysis: how changes in banks' regulatory capital positions affect loan pricing.

Volk (2026) studies corporate lending in Slovenia and finds that banks with unrealized losses equal to 1 percentage point of risk-weighted assets charge, on average, 8 basis points higher lending rates. This effect is smaller than the estimated impact of realized changes in bank capital, which the literature places at around 10–25 basis points (Dagher, Dell’Ariccia, Laeven, Ratnovski, and Tong, 2016). We simulate our baseline economy, in which banks are subject to book-value capital requirements, for 100,000 periods and define banks' unrealized losses as the percentage difference between book-value equity and fair value equity:

$$Unrealized_t = 100 \frac{B_t^{BV} - B_t}{mean(B_t^{BV})}. \quad (47)$$

We define realized capital losses as the shortfall of bank equity relative to its average:

$$Realized_t = 100 \frac{mean(B^{BV}) - B_t^{BV}}{mean(B_t^{BV})} \quad (48)$$

We then estimate the loan-pricing effects of realized and unrealized bank losses using linear regressions of the form

$$spread_t = \beta_0 + \beta_1 Unrealized_t + \beta_2 Realized_t + \beta_3 K_t + \beta_4 Q_t + \beta_5 E_t + \beta_6 \Omega_t^D, \quad (49)$$

where the dependent variable is the spread between the loan rate and the policy rate, measured in annualized percentage points, $spread_t = 400(R_{Lt} - R_t)$. We control for aggregate physical capital and its price, firm equity and profits from deposit taking, $\Omega_t^D = (R_t - R_{Dt})D_t$.

The results are depicted in Figure 1. Consistent with the empirical literature, although somewhat lower in magnitude, unrealized losses on banks' balance sheets have a significant positive effect on loan pricing: $\beta_1 \approx 0.04$, corresponding to an approximately 4 basis point increase in the annualized loan rate. This response is weaker than the effect of realized capital losses, for which $\beta_2 \approx 0.09$, corresponding to an approximately 9 basis point increase in the annualized loan rate.

The intuition is that realized losses directly reduce bank equity and therefore lending capacity, contracting loan supply and increasing the equilibrium loan rate. By contrast, unrealized losses affect loan supply through the shadow value of equity, given by (5). Since

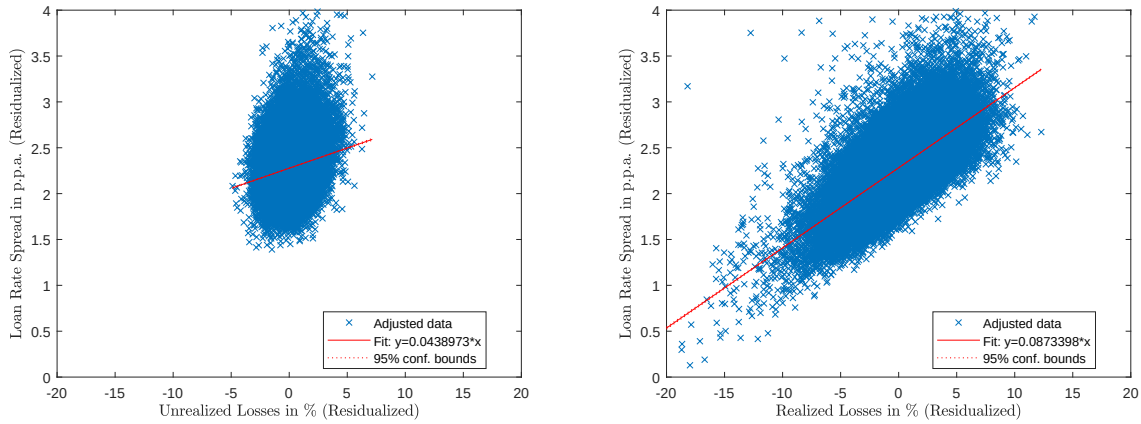


Figure 1: *Loan pricing effects: realized vs. unrealized losses*

Notes: The figure presents coefficients from the linear regression in Equation (49). Variables are residualized with respect to all other regressors using a Frisch–Waugh–Lovell decomposition. The regression is estimated on simulated data from the baseline model.

these losses may be realized in the future with positive probability, they increase the expected scarcity value of equity and thereby raise loan rates even before they are recognized in regulatory capital.

The findings in this section therefore suggest that monetary policy transmission is stronger if banks are required to adjust their regulatory capital for unrealized value changes of bonds portfolio, consistent with empirical evidence in [Greenwald et al. \(2024\)](#) and [Orame et al. \(2025\)](#). The counterfactual exercises in the next section shed more light on this.

4 Regulatory Accounting, Credit Supply and Financial Stability

In this section, we evaluate how including unrealized gains and losses on banks’ long-term bond holdings in regulatory capital affects lending, solvency and broader macroeconomic outcomes. We then draw implications for the optimal accounting rule for measuring regulatory capital. Throughout the analysis, we compare the baseline economy with a counterfactual economy in which regulatory capital incorporates the fair value of banks’ long-term bonds

portfolio, using an identical sequence of exogenous shocks.

4.1 Dynamics Around Episodes with Large Unrealized Losses

Differences between fair-value accounting and book-value accounting become particularly salient when banks have accumulated large unrealized gains or losses on their bond portfolios. These valuation changes are primarily driven by interest rate movements: increases in policy rates reduce the market value of banks' long-term bond holdings, generating unrealized losses.

To study the dynamics of the economy around such episodes, we conduct an event-study exercise based on long simulations of the model. We simulate the baseline economy for 100,000 periods under book-value accounting and construct a parallel simulation under fair-value accounting using the identical sequence of exogenous shocks. Differences in macroeconomic outcomes between the two economies therefore reflect only the regulatory treatment of unrealized gains and losses.

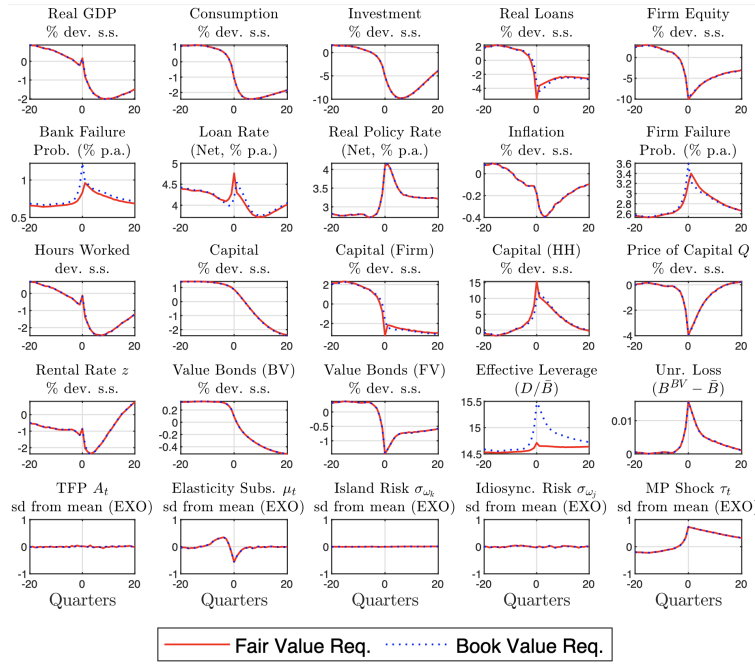
Episodes are identified in the baseline simulation based on the accumulation of unrealized losses on banks' balance sheets. Let accumulated losses be defined as

$$AL_t \equiv B_t^{BV} - B_t,$$

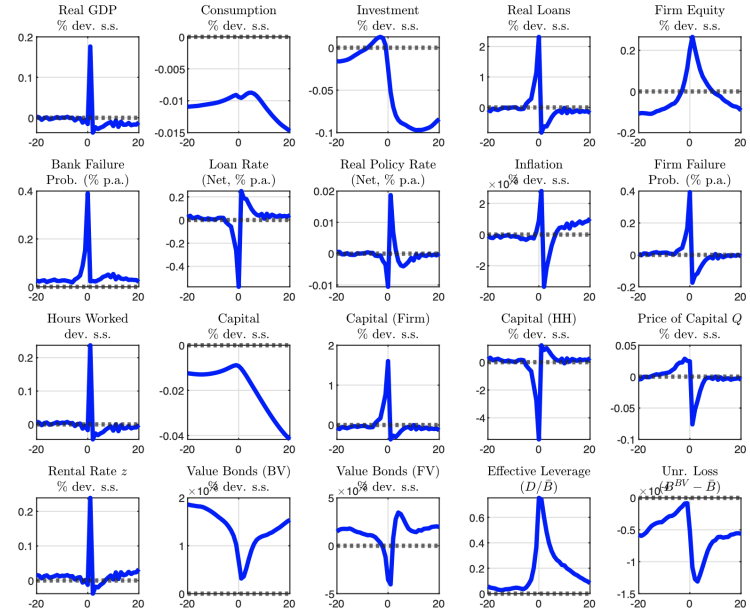
where B_t^{BV} and B_t denote book-value and fair-value equity, respectively. We focus on episodes in which accumulated losses exceed their 90th percentile, denoted by $p_{90}(AL)$. Let $t = 0$ denote the first period of such an episode:

$$AL_{-1} < p_{90}(AL), \quad AL_0 \geq p_{90}(AL).$$

Figure 2 plots the average dynamics of key variables around these events, including both endogenous outcomes and exogenous shocks. Episodes with large accumulated unrealized losses are typically preceded by a period of monetary easing, during which low policy rates raise bond valuations. Over the same period, a temporary increase in the elasticity of substitution across intermediate-goods producers—that is, a decline in cost-push pressures—builds gradually and peaks ahead of the event, contributing to low inflation and low real interest rates.



(a) Levels



(b) Differences (Book Value - Fair Value)

Figure 2: Anatomy: Episodes with large accumulated unrealized losses

Notes: Time 0 denotes the start of a spell in which banks' balance sheet contain large accumulated unrealized losses, such that $AL_{-1} < p_{90}(AL)$ and $AL_0 \geq p_{90}(AL)$, where $AL_t \equiv B_t^{BV} - B_t$. Exogenous variables are labeled as EXO.

Shortly before the event, both forces begin to reverse. The unwinding of monetary accommodation, reinforced by rising cost-push pressures, generates a sharp increase in the real policy rate, which peaks at the event. This abrupt tightening leads to a large decline in bond valuations, turning earlier valuation gains into accumulated unrealized losses. The resulting dynamics are broadly reminiscent of the post-COVID tightening cycle, which was characterized by sharp monetary tightening partly in response to cost-push pressures.

The evolution of loan supply and bank solvency concerns during these episodes depends on the regulatory accounting framework. Under the book-value regime, banks are able to maintain a higher supply of loans while bond prices are falling. Because unrealized losses are not reflected in regulatory capital, banks effectively operate with higher leverage. As a result, bank default probabilities gradually rise relative to the fair-value regime during the monetary tightening phase.

During the subsequent normalization of monetary policy, when bond prices begin to recover, loan supply becomes higher under fair-value accounting. Two mechanisms contribute towards this. First, the fair value of bank equity responds more quickly to changes in bond prices than its book-value counterpart, allowing regulatory capital to recover sooner. Second, the sustained increase in bank default probabilities under book-value accounting during the tightening phase generates larger realized equity losses, resulting in lower lending during the subsequent policy normalization phase.

4.2 Valuation Effects on Lending and Bank Defaults

We now turn to a closer examination of how the prudential treatment of unrealized gains and losses affects credit supply and bank default probabilities. To do so, we exploit the two parallel simulations described above—one under book-value accounting and one under fair-value accounting—constructed using the same sequence of exogenous shocks. In the baseline simulation, we measure unrealized losses on banks’ balance sheets as the percentage gap between book-value and fair-value equity, $(B_t^{BV} - B_t)/B_t^{BV}$, so that positive values correspond to unrealized losses and negative values to unrealized gains. For each realization of this measure, we compare loan quantities and bank default probabilities across the two simulations. This allows us to evaluate the two regulatory regimes at identical realizations of the underlying shocks.

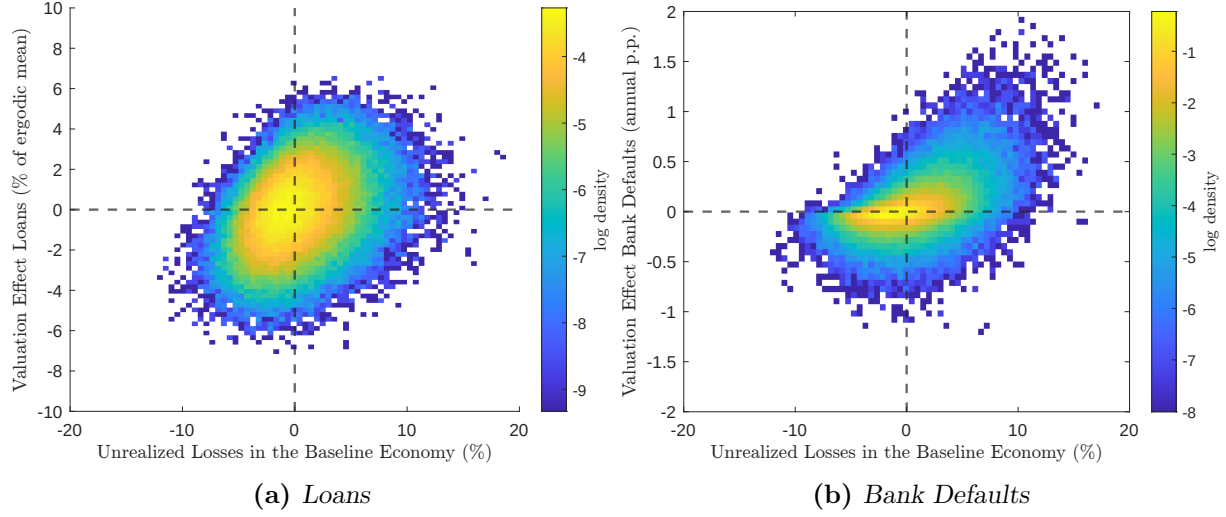


Figure 3: Valuation Effects

Notes: The left panel of this figure shows the Valuation Effect on Loans, defined as the difference in bank loans between the baseline economy (book-value regime) and the alternate economy (fair-value regime) for different levels of unrealized losses or gains. The difference in bank loans is reported as a percentage of the ergodic mean of loans under the book-value regime. The right panel of this figure shows the Valuation Effect on Bank Defaults, defined as the difference in the annualized bank default probability between the baseline economy (book-value regime) and the alternate economy (fair-value regime) for different levels of unrealized losses or gains. The color scale indicates the log-density of observations in the simulated data.

We summarize the effects of the regulatory valuation regime using two measures. The left panel of Figure 3 reports the *Valuation Effect on Loans*, defined as lending under book-value accounting minus lending under fair-value accounting, evaluated at different levels of unrealized gains or losses on banks' bond portfolios. The difference in lending is expressed as a percentage of the ergodic mean of loans under the book-value regime. The right panel reports the *Valuation Effect on Bank Defaults*, defined as the annualized probability of bank default under book-value accounting minus the corresponding probability under fair-value accounting.

The figure shows that when unrealized losses (gains) accumulate on banks' balance sheets, both lending and bank default probabilities tend to be higher (lower) under book-value accounting.

Figure 4 highlights the close, but imperfect, relationship between the two effects. Periods in which lending is higher under the book-value regime also tend to be periods in which

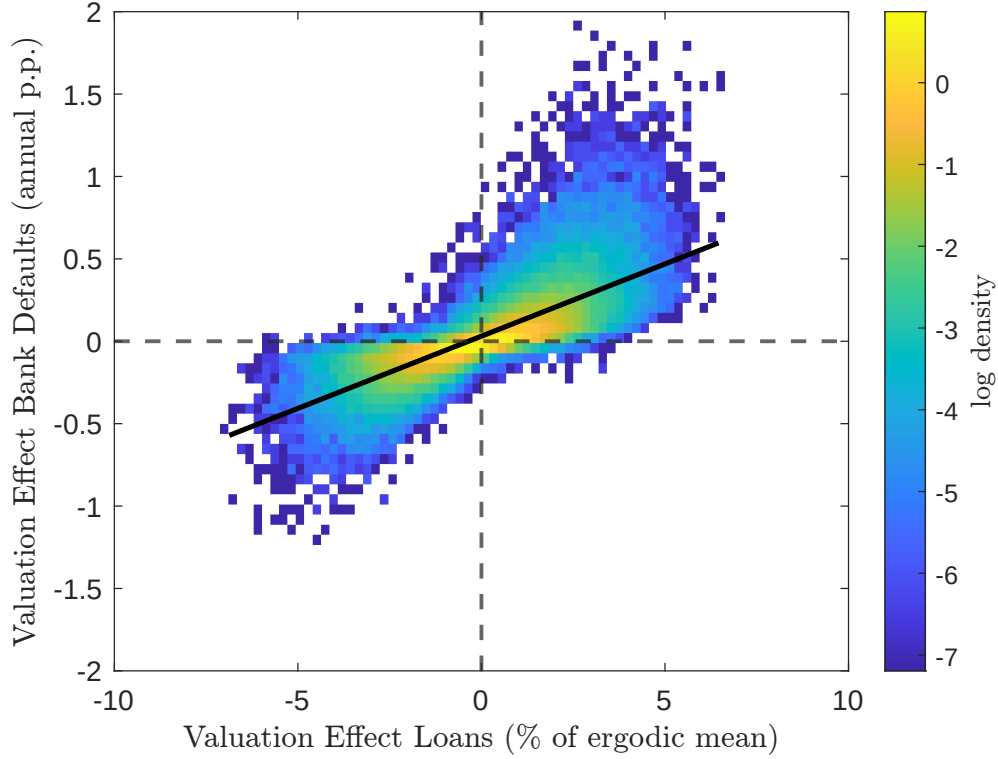
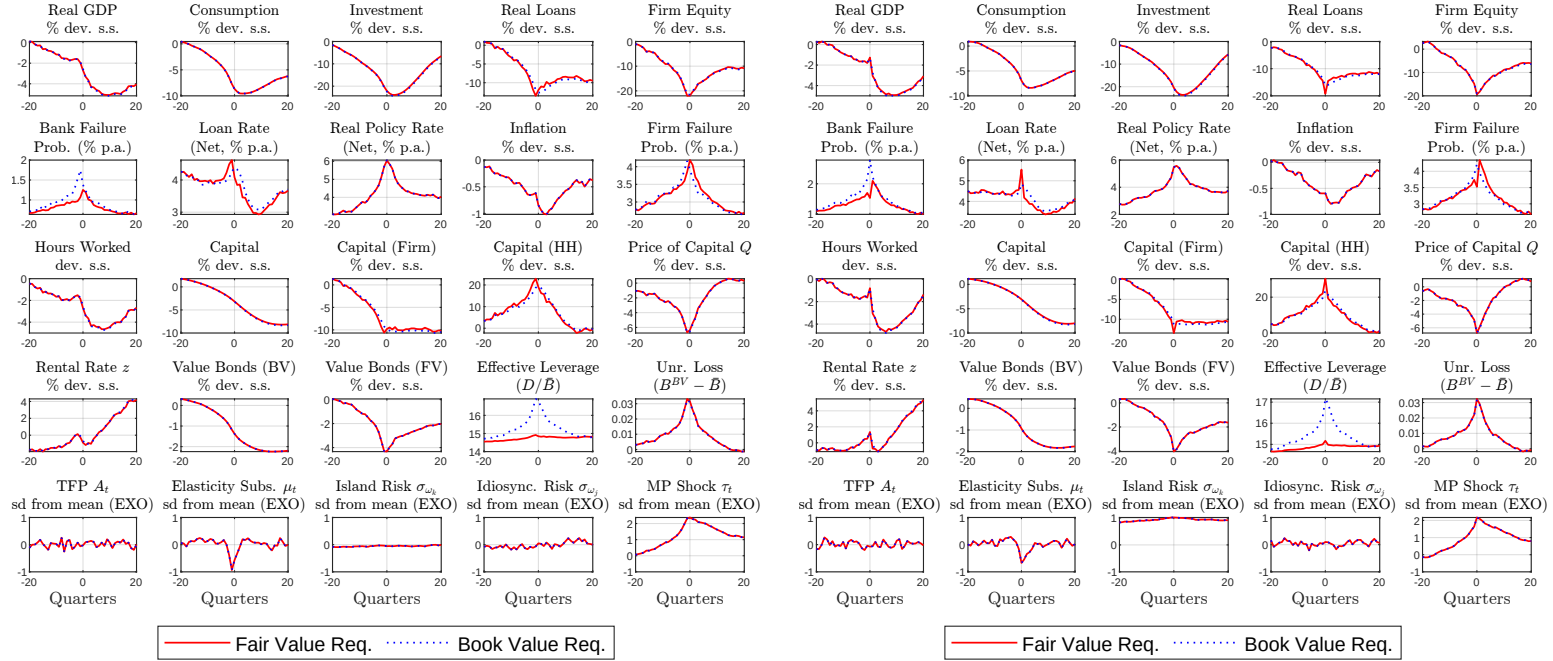


Figure 4: *Valuation Effects on Lending and Bank Defaults: Correlation*

Notes: The heatmap shows the log-density of simulated observations. The solid line represents a linear fit across all points. The Valuation Effect on Loans is defined as the difference in bank loans between the baseline economy (book-value regime) and the alternate economy (fair-value regime) for different levels of unrealized losses or gains. The difference in bank loans is reported as a percentage of the ergodic mean of loans under the book-value regime. The Valuation Effect on Bank Defaults is defined as the difference in the annualized bank default probability between the baseline economy (book-value regime) and the alternate economy (fair-value regime) for different levels of unrealized losses or gains.

bank default probabilities increase the most. The mechanism behind this relationship is straightforward: because unrealized losses do not reduce regulatory capital under book-value accounting, banks face a looser capital constraint and are able to extend more loans. The resulting increase in effective leverage raises the probability of bank insolvency.

These effects are economically meaningful. The average annual bank failure probability in the book-value model is 0.7%. For unrealized losses equal to 4% of regulatory equity—losses at least this large occur in 10% of periods—the median valuation effect on bank defaults amounts to about 0.11 percentage points.



(a) Low valuation effect on bank defaults

(b) High valuation effect on bank defaults

Figure 5: Anatomy: The role of credit risk

Notes: The Valuation Effect on Bank Defaults is defined as the difference in the annualized bank default probability between the baseline economy (book-value regime) and the alternate economy (fair-value regime). The effect is labeled low (high) when it is in its first (fourth) quartile. Time 0 denotes the start of a spell in which banks' balance sheet contain large accumulated unrealized losses, such that the fair value of equity is between 10 and 30% below it's book value. Exogenous variables are labeled as EXO.

At the same time, the valuation effects exhibit substantial dispersion. Even for similar levels of accumulated unrealized losses, the impact on lending and bank default probability varies widely across observations. This suggests that differences in underlying macroeconomic and financial conditions matter for assessing the impact of regulatory accounting treatment of bond valuation changes.

To understand the source of this dispersion, we examine event windows around episodes with large accumulated unrealized losses, splitting them by the magnitude of the valuation effect on bank defaults. Specifically, we consider periods in which the fair value of bank equity is between 10 and 30% below its book value. The left panel of Figure 5 shows the evolution of the economy when the valuation effect on bank defaults lies in the first quartile of its distribution, while the right panel shows episodes in the fourth quartile.

The differences across the two panels are largely driven by variation in the level of credit risk. In the model, credit risk arises from firms' exposure to island-specific shocks that banks cannot diversify away. When the variance of these shocks is high (right panel), bank default probabilities increase substantially regardless of the accounting regime. In such high-risk environments, however, the higher effective leverage permitted under book-value regime becomes particularly detrimental to bank solvency.²⁸

4.2.1 The role of credit risk

Motivated by the differences in Figure 5, we turn to study the implications that banks' exposure to credit risk has for the valuation effects over the monetary policy cycle. Table 3 reports average outcomes of relevant variables conditional on the monetary policy regime and the level of credit risk. We classify periods as monetary tightening (easing) episodes if the sign of the monetary policy shock is positive (negative). We further classify periods as high- or low-risk states based on the variance of island risk shocks, with high-risk states defined as periods in which the variance is above the 75th percentile and low-risk states as periods in which it is below the median.

Several patterns are noteworthy. Most prominently, the magnitude of the valuation effect on bank defaults is asymmetric over the monetary policy cycle. Column 1 shows that, during tightening periods, the average bank default probability under the book-value regime is

²⁸Begenau et al. (2025b) show that the 2022-23 banking crisis in the U.S. reflects such an environment, when a high realization of interest rate risk coincided with a high realization of credit risk.

Table 3: Average Outcomes by Accounting Regime, Monetary Policy, and Credit Risk

	Tightening	Easing	Tightening_HighRisk	Tightening_LowRisk	Easing_HighRisk	Easing_LowRisk
Loans (FV)	3.8258	4.322	3.5346	4.0062	3.9837	4.5168
Loans (BV)	3.8287	4.3156	3.5358	4.0099	3.977	4.511
Loans (BV - FV)	0.0028	-0.0064	0.0012	0.0037	-0.0066	-0.0058
Bank Defaults (FV)	0.8529	0.4394	1.6321	0.4152	0.9724	0.1771
Bank Defaults (BV)	0.8961	0.4381	1.7017	0.4428	0.9632	0.1792
Bank Defaults (BV - FV)	0.0432	-0.0013	0.0696	0.0276	-0.0093	0.0021
Firm Defaults (FV)	3.171	2.0635	3.4382	3.0324	2.3989	1.8965
Firm Defaults (BV)	3.1697	2.0814	3.4429	3.0274	2.4144	1.9154
Firm Defaults (BV - FV)	-0.0013	0.0179	0.0047	-0.005	0.0155	0.0189
Real Loan Spread p.p.a (FV)	2.5579	1.9927	2.8491	2.3953	2.2108	1.895
Real Loan Spread p.p.a (BV)	2.5537	2.0277	2.8527	2.3879	2.2438	1.9293
Real Loan Spread p.p.a (BV - FV)	-0.0042	0.035	0.0036	-0.0073	0.0329	0.0343
Policy Rate p.p.a. dev. (FV)	0.7663	-0.762	0.8072	0.7414	-0.7254	-0.7748
Policy Rate p.p.a. dev. (BV)	0.7659	-0.7616	0.807	0.7408	-0.725	-0.7744
Policy Rate p.p.a. dev (BV - FV)	-0.0004	0.0004	-0.0003	-0.0006	0.0004	0.0004
Unrealized Losses (FV)	0.0047	-0.0039	0.0049	0.0045	-0.0038	-0.0038
Unrealized Losses (BV)	0.0046	-0.0039	0.0048	0.0045	-0.0038	-0.0039
Unrealized Losses (BV - FV)	-0.0001	-0.0001	-0.0001	0	0	-0.0001
Frequency Unr. Losses > 0 (%)	68.5305	29.3365	69.1439	68.1601	29.7578	29.9393
Frequency Regime (%)	49.8584	50.1406	12.677	24.4048	12.3227	25.5947

Note: This table reports averages over a simulation of 100,000 periods. Tightening episodes are defined as those in which the policy rate is above the one implied by the Taylor Rule, i.e. $\tau_t > 1$, and easing episodes as those in which $\tau_t < 1$. Stability episodes are those in which the variance of the island-specific shock σ_{ω_k} is particularly low (below its median), and instability episodes those with σ_{ω_k} above its 75th percentile.

0.04 percentage points, or about 5%, higher than under the fair-value regime. During easing periods, the corresponding difference is negative but economically not meaningful. The reason for this asymmetry is that firm default probabilities and long-term bond prices tend to negatively co-move over the cycle. During tightening episodes, rising policy rates both generate unrealized losses on long-term bonds and contribute to higher firm default probabilities, which in turn raises bank default probabilities.²⁹ By constraining bank leverage during these periods, fair-value accounting has substantial stabilizing effects. As seen in column 3, the magnitude of the stabilizing effect is therefore largest when monetary policy tightens in the high-credit-risk state. During easing episodes, by contrast, valuation gains relax regulatory constraints in the fair-value regime when borrower default risk is lower, so the corresponding increase in bank default probability is negligible (see column 2). When monetary policy loosens in the low-credit-risk state, bank default probability is in fact lower in the fair-value regime. This is because higher lending by banks in these periods is particularly profitable

²⁹Figure 9 in the Online Appendix shows that the 12-quarter rolling correlation between firm default probabilities and unrealized bond valuation losses in our estimated model is predominantly positive.

thanks to the much lower risk of borrower defaults (see column 6).

The asymmetric effect on bank solvency shapes the response of credit spreads and aggregate credit supply. Under book-value-based capital requirements, the higher lending capacity of banks due to relatively looser lending constraints during monetary tightening is largely offset by higher bank failures and the resulting losses in aggregate bank capital. Fair value-based requirements therefore produce only a moderately larger contraction in credit, despite imposing tighter regulatory constraints. During easing episodes, the additional lending capacity created by recognition of valuation gains under fair-value accounting is only partially offset by the modest increase in bank failures, leading to a positive impact on credit which is larger in magnitude relative to the negative impact during monetary tightening. The asymmetric impact on loan spreads is even more pronounced than that on lending due to the relatively inelastic aggregate loan demand.³⁰

These mechanisms help explain the imperfect relationship between the valuation effects on lending and bank defaults observed in Figure 4. Figure 6 shows this directly by drawing scatter plots of the two valuation effects conditional on the monetary policy regime and level of credit risk. For illustrative purposes, we use a stricter definition for classifying the monetary policy regime where in addition to the sign of the monetary policy shock, we require that the real policy rate is rising (strict tightening) or falling (strict easing).³¹

The relationship is particularly flat when easing episodes coincide with the low-credit-risk state. In such an environment, the relatively looser regulatory constraint in the fair-value regime stimulates greater lending, while the impact on bank default probability remains limited due to the especially low risk of borrower defaults. The relationship becomes stronger in the high-credit-risk state, since higher lending in the fair-value regime is accompanied by high firm defaults which, in turn, increases the bank default probability. During tightening episodes, when the constraint is relatively looser in the book-value regime, the relationship

³⁰As discussed in the model section, aggregate credit supply is perfectly inelastic as long as the loan rate is high enough for banks to prefer lending up to the regulatory constraint: it is fixed by the scarcity of (regulatory) bank equity. Hence, differences in loan rates reflect movements along (and shifts of) the aggregate loan demand curve.

³¹The second-to-last row in Table 3 shows that unrealized gains or losses can occur in both monetary policy regimes under our baseline classification. This is largely due to the effect of past shocks. For this same reason, the real policy rate—and their corresponding effect on bond prices—could also be decreasing or increasing in either regimes. Figure 6 therefore focuses on the “stricter” classification. For completeness, Figure 8 in the Appendix reproduces the plot using our baseline classification of monetary policy regimes.

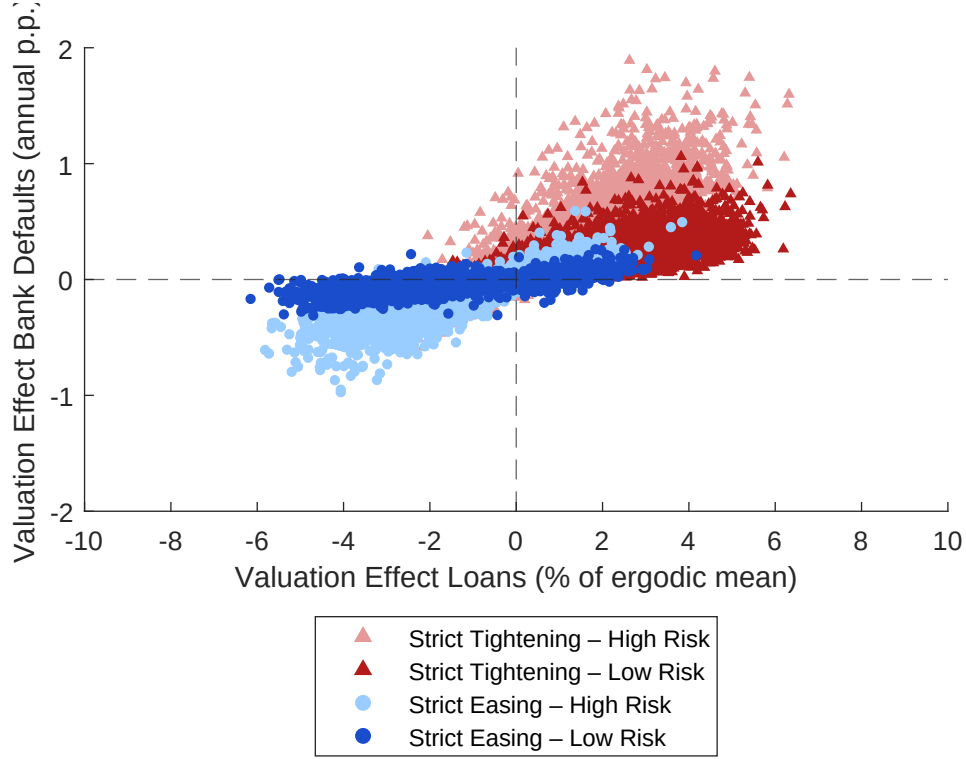


Figure 6: *Valuation Effects on Lending and Bank Defaults: Correlation by Regime*

Notes: Strict tightening episodes are defined as those in which the real policy rate is above the one implied by the Taylor Rule, i.e. $\tau_t > 1$ and *increasing*, while strict easing episodes as those in which $\tau_t < 1$ and the real policy rate is *decreasing*. Stability episodes are those in which the variance of the island-specific shock σ_{ω_k} is particularly low (below its median), and instability episodes those with σ_{ω_k} above its 75th percentile. The Valuation Effect on Loans is defined as the difference in bank loans between the baseline economy (book-value regime) and the alternate economy (fair-value regime) for different levels of unrealized losses or gains. The difference in bank loans is reported as a percentage of the ergodic mean of loans under the book-value regime. The Valuation Effect on Bank Defaults is defined as the difference in the annualized bank default probability between the baseline economy (book-value regime) and the alternate economy (fair-value regime) for different levels of unrealized losses or gains.

is instead weaker when credit risk is high since realized equity losses from high default rates dampen lending capacity (notice the much larger dispersion in the valuation effect on bank defaults for a given effect on lending).

In summary, the effects of regulatory accounting treatment of bond valuation changes on lending and bank solvency are highly asymmetric over the monetary policy cycle. The next subsection discusses the implications of this asymmetry for welfare comparisons between the

book-value and fair-value regulatory regimes.

4.3 Welfare-Maximizing Regulatory Accounting Approach

The previous subsections show that regulatory accounting affects credit supply and financial stability in a state-dependent manner. In particular, fair-value accounting changes both the response of lending to monetary policy and the frequency of bank failures. These mechanisms translate into differences in aggregate consumption through two channels: lending affects capital accumulation and production, while bank defaults generate deadweight losses that reduce resources available for consumption.

We now evaluate how these mechanisms affect social welfare and identify the regulatory accounting framework that maximizes welfare in our model. We assess both whether regulatory capital should be defined on the basis of the book value or the fair value of equity, and how high the capital requirements should be. We find that regulatory capital defined on the basis of the fair-value of the bonds is overall superior in terms of welfare to defining it on the basis of their book value.

Table 4: *Summary Statistics by Accounting Regime*

Variable	Mean		SD	
	FV	BV	FV	BV
Loans (L_t)	4.0746	4.0728	0.4075	0.4065
Real Loan Spread p.p.a $\left[400 \times \frac{(R_{Lt}-R_t)}{\Pi_{t+1}}\right]$	2.2745	2.29	0.9671	0.9069
Bank Defaults p.a. $[400 \times F_{kt}(\bar{\omega}_{bt})]$	0.6455	0.6664	0.6281	0.6646
Firm Defaults p.a $[400 \times \int_0^\infty F_{jt}(\bar{\omega}_{jt}(\omega_k))dF_{kt}]$	2.6157	2.624	0.9417	0.9536
Consumption (C_t)	1.0499	1.0497	0.047	0.047
Total Production (Y_t)	1.6958	1.6958	0.0645	0.0646
Utility Difference (% CE)	0.0147			

Note: The table reports the means and standard deviations (SD) of selected variables under the fair-value (FV) and book-value (BV) accounting regime. The utility difference is expressed in consumption equivalent terms, see Appendix C.

Table 4 reports means and standard deviations of important financial and real economy variables by accounting regime. Additionally, Figure 7 shows how selected variables change

with the level of the capital requirement in each of the regimes.

On average, banks supply slightly higher loan volumes at slightly lower spreads under fair value requirements. The default probability of both banks and firms is also reduced.

These average outcomes reflect the state-dependent mechanisms documented in the previous subsection and their relative size. In particular, the effect of the accounting regime on bank defaults is much stronger during monetary tightening episodes, when policy rate increases generate unrealized losses on banks' bond portfolios. At the same time, the effect on lending is larger during easing episodes. This asymmetry implies that bank default probabilities are on average lower and lending is higher under fair-value accounting. The lower firm default probability under fair-value accounting is then a consequence of lower aggregate loan rates. This effect is reinforced by the feedback loop between lower loan rates, lower firm default probabilities and lower bank default probabilities inherent in the model.

The on average lower default probabilities of firms and banks under fair-value accounting reduce deadweight default costs, generating consumption gains. At the same time, on average larger credit supply under fair-value accounting supports investment and production, also generating consumption gains. Thus, consumption is overall higher under fair-value accounting, which translates into welfare gains.

Our results differ from those in [Acharya et al. \(2021\)](#), who analyze accounting regimes in a setting with asymmetric information about loan quality and creditor runs. In their framework, fair-value accounting improves ex-ante investment efficiency by revealing information about project quality but may increase financial fragility through belief dispersion and run incentives. In contrast, our model abstracts from asymmetric information and focuses on the regulatory treatment of unrealized valuation changes on relatively safe liquid assets. In this environment, recognizing unrealized losses primarily affects bank leverage and the incidence of costly bank failures rather than investors' information about asset quality. As a result, fair-value accounting in our framework tends to reduce bank failures in states where valuation losses are large, which contributes to the welfare gains we document.

The quantitative welfare differences between the two accounting regimes are modest. For an 8% capital requirement, the welfare gains from fair-value accounting amount to approximately 1.5 basis points in consumption-equivalent terms.

Finally, we also evaluate the optimal level of capital requirements. [Figure 7](#) shows that welfare is maximized at a capital ratio of approximately 12%, which is substantially higher

than the Basel II minimum but below the 16% level found in [Mendicino et al. \(2026\)](#). Optimal fair-value capital requirements of 12% correspond to an increase in welfare equivalent to a roughly 13.6 bps increase in consumption equivalent utility terms.

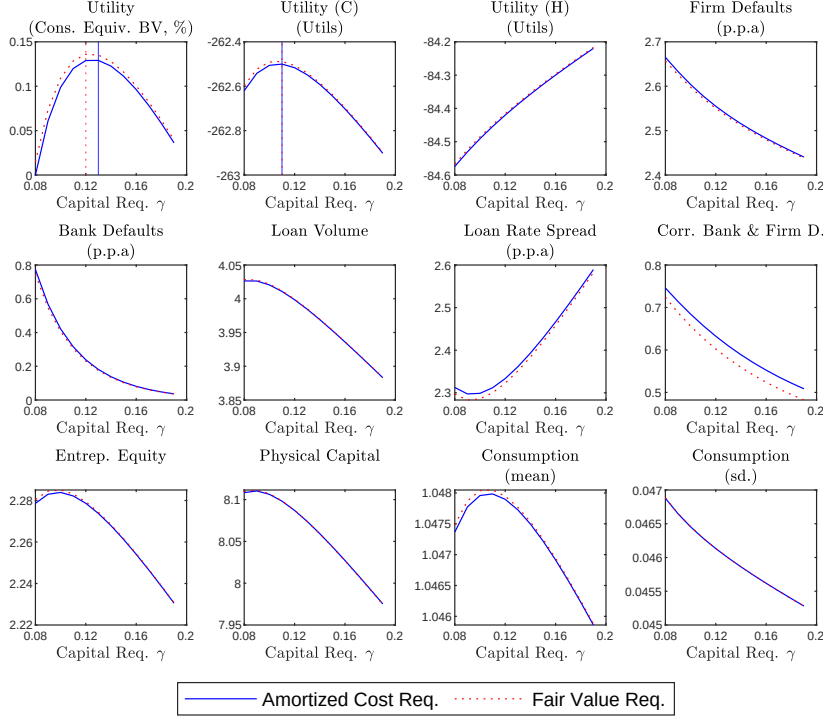


Figure 7: *Optimal capital requirements*

Notes: Utility is household utility, which is the relevant welfare benchmark as all banks and firms are owned by household. Utility (C) and Utility (H) are the parts attributed to consumption and labor, respectively. The benchmark for computing consumption equivalents is the baseline model with a capital requirement of 8%. All parameters other than the capital requirement γ are kept fixed.

Taken together, these results highlight the importance of the regulatory treatment of unrealized gains and losses for both financial stability and credit supply. While fair-value accounting tightens regulatory constraints during periods of monetary tightening, it substantially reduces bank failures in precisely those states in which credit risk is elevated. Because the reduction in costly defaults during tightening episodes dominates the increase during easing episodes—and because lending expands more strongly under fair-value accounting when monetary policy eases—the overall effect is higher average consumption and welfare

under fair-value regulatory capital. The welfare-maximizing policy in our model therefore combines the recognition of unrealized gains and losses in regulatory capital with capital requirements that are higher than those prevailing under Basel II.

5 Conclusion

This paper examines how the regulatory capital treatment of unrealized capital gains and losses on banks' debt securities, related to the impact of interest rate risk, affects financial stability and credit supply. To this purpose, it develops a dynamic general equilibrium model in which banks are exposed to both interest rate risk and credit risk.

Our results highlight that the effects of regulatory capital accounting are inherently state-dependent, giving rise to pronounced asymmetries over the monetary policy cycle. The key mechanism is that credit risk amplifies the impact of accounting rules on bank solvency. When underlying credit risk is elevated, differences across accounting regimes translate into much larger differences in bank default probabilities. As a result, book-value accounting is particularly detrimental to bank solvency during monetary tightening, when higher interest rates both reduce bond valuations and increase borrower default risk. During easing episodes, the same forces operate in reverse, and the implications of the accounting regime for bank solvency are correspondingly more muted.

These asymmetries in solvency outcomes also shape the response of lending. In tightening episodes, the higher incidence of bank distress under book-value accounting erodes bank equity and partially offsets the looser regulatory constraints implied by that regime, limiting its relative effect on credit supply. By contrast, in easing periods, when solvency concerns are less acute, differences in regulatory constraints translate more directly into lending outcomes.

Thus, on average, fair-value accounting involves a lower bank default probability and more lending. Having regulatory capital defined on the basis of the fair-value of the bonds is overall slightly superior in terms of welfare to having regulatory capital defined on the basis of their book value. Hence, our results lend support to the inclusion of accumulated other comprehensive income (AOCI), which records valuation changes on Available-for-Sale assets, in the definition of tier 1 capital of Basel III.

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Appendix

A Model Details

This Appendix describes elements of the standard New Keynesian models omitted in Section 2 of the main text, and presents the details of the equilibrium.

A.1 Household Problem

Households maximize Equation (1) subject to the budget constraint Equation (2). We begin by providing details of the components of household's cash flows, summarized by Σ_t in Equation (2). Households:

- receive net payoffs $(1 - \theta_B)(1 - \xi_B)\rho_t^B P_{t-1}B_{t-1}$ from bankers,
- receive net payoffs $(1 - \theta_E)(1 - \xi_E)\rho_t^E P_{t-1}E_{t-1}$ from entrepreneurs,
- receive $(P_t - P_t^m)Y_t - \frac{\theta_R}{2}(\Pi_t - 1)^2 P_t Y_t$ from final good producers (see Section A.3),
- receive profits $P_t \Pi_t^C$ from capital producers and $P_t(\varsigma_t K_t^H - \frac{\kappa_H}{2}(K_t^H)^2)$ from capital managers (see Section A.4),
- are charged a lump sum fee $\frac{\theta_W}{2}(\Pi_t^W - 1)^2 W_t$ from a labor union that negotiates wages (explained in the next subsection),
- are charged lump sum taxes LT_t by the government.

Therefore:

$$\begin{aligned} \Sigma_t = & (1 - \theta_B)(1 - \xi_B)\rho_t^B P_{t-1}B_{t-1} + (1 - \theta_E)(1 - \xi_E)\rho_t^E P_{t-1}E_{t-1} + P_t(\varsigma_t K_t^H - \frac{\kappa_H}{2}(K_t^H)^2) \\ & + P_t \Pi_t^C + (P_t - P_t^m)Y_t - \frac{\theta_R}{2}(\Pi_t - 1)^2 P_t Y_t - \frac{\theta_W}{2}(\Pi_t^W - 1)^2 W_t - LT_t. \end{aligned} \quad (\text{A.1})$$

The multiplier on the budget constraint Equation (2) (which is expressed in nominal terms) is $\tilde{\lambda}_t$. We can then define the multiplier on the real budget constraint as $\lambda_t = P_t \tilde{\lambda}_t$ and

express the real stochastic discount factor as

$$\Lambda_{t,t+1} = \beta \frac{\lambda_{t+1}}{\lambda_t}. \quad (\text{A.2})$$

We obtain the first order conditions for consumption, capital, deposits, and riskless one-period bonds, as follows:

$$(C_t - bC_{t-1})^{-\sigma} - \beta b \mathbb{E}_t(C_{t+1} - bC_t)^{-\sigma} = \lambda_t, \quad (\text{A.3})$$

$$\mathbb{E}_t \Lambda_{t,t+1} [z_{t+1} + (1 - \delta)Q_{t+1}] = Q_t + \varsigma_t, \quad (\text{A.4})$$

$$\mathbb{E}_t \Lambda_{t,t+1} \frac{R_{Dt}}{\Pi_{t+1}} = 1. \quad (\text{A.5})$$

As explained in the main text, the labor supply decision is relegated to a labor union, whose problem is described in continuation.

A.2 Wage Setting

Wage setting is subject to [Rotemberg \(1982\)](#) adjustment costs governed by parameter θ_W which the labor union finances by charging households a lump-sum fee. No costs arise from adjusting wages according to the steady-state inflation $\bar{\Pi}$.³² The labor packer's demand for variety h is:

$$H_{ht} = \left(\frac{W_{ht}}{W_t} \right)^{-\epsilon_W} H_t. \quad (\text{A.6})$$

The labor union maximizes household utility subject to labor demand (Equation A.6 with Lagrange multiplier mrs_t) and the nominal household budget constraint (Eq. 2 with multiplier λ_t/P_t):

$$\begin{aligned} \max_{H_{ht}, W_{ht}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t & \left[\lambda_t \left(\frac{W_{ht}}{P_t} H_{ht} - \frac{\theta_W}{2} \left(\frac{W_{ht}}{\bar{\Pi} W_{ht-1}} - 1 \right)^2 \frac{W_t}{P_t} \right) \right. \\ & \left. - \frac{\xi_H H_{ht}^{1+\varphi_H}}{1+\varphi_H} - mrs_t \left(H_{ht} - \left(\frac{W_{ht}}{W_t} \right)^{-\epsilon_W} H_t \right) \right]. \end{aligned} \quad (\text{A.7})$$

³²This may be thought of as wages being fully indexed to steady-state inflation: If the union does not actively adjust nominal wages, they grow by the steady-state inflation rate.

We obtain the first order conditions for hours worked and wages as follows:

$$mrs_t = \lambda_t \frac{W_{ht}}{P_t} - \xi_H H_{ht}^{\varphi_H}, \quad (\text{A.8})$$

$$\begin{aligned} \theta_W \lambda_t \left(\frac{W_{ht}}{\bar{\Pi} W_{ht-1}} - 1 \right) \frac{W_t}{P_t} \frac{1}{\bar{\Pi} W_{ht-1}} &= \lambda_t \frac{H_{ht}}{P_t} - \epsilon_W mrs_t \left(\frac{W_{ht}}{W_t} \right)^{-\epsilon_W - 1} \frac{H_t}{W_t} \\ &\quad - \beta \mathbb{E}_t \left[\lambda_{t+1} \theta_W \left(\frac{W_{ht+1}}{\bar{\Pi} W_{ht}} - 1 \right) \frac{W_{t+1}}{P_{t+1}} \frac{W_{ht+1}}{\bar{\Pi} W_{ht}^2} \right]. \end{aligned} \quad (\text{A.9})$$

Since the problem is identical for each variety h , there is no price dispersion and hence $W_{ht} = W_t$. Define the nominal wage inflation as $\Pi_t^W = \frac{W_t}{W_{t-1}}$. Then, the first order conditions can be combined to obtain the New Keynesian Wage Phillips Curve:

$$\theta_W \left(\frac{\Pi_t^W}{\bar{\Pi}} - 1 \right) \frac{\Pi_t^W}{\bar{\Pi}} = \theta_W \mathbb{E}_t \left[\Lambda_{t,t+1} \left(\frac{\Pi_{t+1}^W}{\bar{\Pi}} - 1 \right) \frac{(\Pi_{t+1}^W)^2}{\Pi_{t+1} \bar{\Pi}} \right] + (1 - \epsilon_W) H_t + \epsilon_W \frac{\xi_H H_t^{1+\varphi_H}}{\lambda_t w_t}, \quad (\text{A.10})$$

where

$$w_t = \frac{w_{t-1} \Pi_t^W}{\Pi_t} \quad (\text{A.11})$$

is the law of motion of real wages.

A.3 Production

A representative competitive intermediate good producer has access to Cobb-Douglas production technology:

$$Y_t^m = \theta_t K_{t-1}^\alpha H_t^{1-\alpha}, \quad (\text{A.12})$$

where $\alpha \in [0, 1]$, and θ_t is an aggregate productivity shock that evolves according to an AR(1) process:

$$\log(\theta_t) = \rho_\theta \log(\theta_{t-1}) + \sigma_\theta \epsilon_{\theta t}, \quad (\text{A.13})$$

with $\epsilon_{\theta t} \sim N(0, 1)$. Denote the price of the intermediate good by P_t^m . The profit maximization problem of the representative intermediate good producer is then:

$$\max_{H_t, K_{t-1}} P_t^m \theta_t K_{t-1}^\alpha H_t^{1-\alpha} - P_t z_t K_{t-1} - P_t w_t H_t. \quad (\text{A.14})$$

Further, define the real price of the intermediate good. as $\frac{P_t^m}{P_t} = mc_t$. Then, the profit maximization problem yields the following FOCs:

$$mc_t \alpha \frac{Y_t^m}{K_{t-1}} = z_t \quad (\text{A.15})$$

$$mc_t (1 - \alpha) \frac{Y_t^m}{H_t} = w_t \quad (\text{A.16})$$

To incorporate nominal price rigidities, we model a unit continuum of monopolistic final good producers, each producing a differentiated variety i using a linear technology with the intermediate good as the only input:

$$Y_{it} = Y_t^m(i). \quad (\text{A.17})$$

The final good composite is the CES aggregate:

$$Y_t = \left(\int_0^1 Y_{it}^{\frac{\mu_t-1}{\mu_t}} di \right)^{\frac{\mu_t}{\mu_t-1}}, \quad (\text{A.18})$$

where μ_t is a stochastic elasticity of substitution that follows an AR(1) process:

$$\ln(\mu_t) = (1 - \rho_\mu) \ln(\mu) + \rho_\mu \ln(\mu_{t-1}) + \sigma_\mu \epsilon_{\mu t}, \quad (\text{A.19})$$

where $\epsilon_{\mu t} \sim N(0, 1)$ is an exogenous markup shock. Final good producers are subject to Rotemberg adjustment costs, governed by parameter θ_R , from steady state inflation $\bar{\Pi}$. They discount real payoffs by the household's real discount factor $\beta^t \lambda_t$. Their maximization

problem in real terms is then:

$$\max_{P_t(i)} \mathbb{E}_t \sum_{n=0}^{\infty} \beta^n \frac{\lambda_{t+n}}{\lambda_t} \left[\frac{P_{t+n}(i)}{P_{t+n}} Y_{it+n} - mc_{t+n} Y_{it+n} - \frac{\theta_R}{2} \left(\frac{P_{t+n}(i)}{\bar{\Pi} P_{t+n-1}(i)} - 1 \right)^2 Y_{t+n} \right] \quad (\text{A.20})$$

$$\text{s.t. } Y_{it} = \left(\frac{P_t(i)}{P_t} \right)^{-\mu_t} Y_t \quad (\text{A.21})$$

The first order condition is:

$$\begin{aligned} (1 - \mu_t) \left(\frac{P_t(i)}{P_t} \right)^{-\mu_t} \frac{Y_t}{P_t} + mc_t \mu_t \left(\frac{P_t(i)}{P_t} \right)^{-\mu_t-1} \frac{Y_t}{P_t} - \theta_R \left(\frac{P_t(i)}{\bar{\Pi} P_{t-1}(i)} - 1 \right) \frac{Y_t}{\bar{\Pi} P_{t-1}(i)} \\ + \beta \mathbb{E}_t \frac{\lambda_{t+1}}{\lambda_t} \theta_R \left(\frac{P_{t+1}(i)}{\bar{\Pi} P_t(i)} - 1 \right) \frac{P_{t+1}}{\bar{\Pi} P_t^2} Y_{t+1} \stackrel{!}{=} 0. \end{aligned} \quad (\text{A.22})$$

Since all final good producers face identical marginal costs, all charge the same price. Therefore the i index can be dropped and one obtains the New Keynesian Phillips Curve:

$$\theta_R \left(\frac{\Pi_t}{\bar{\Pi}} - 1 \right) \frac{\Pi_t}{\bar{\Pi}} = \mathbb{E}_t \Lambda_{t,t+1} \theta_R \left(\frac{\Pi_{t+1}}{\bar{\Pi}} - 1 \right) \frac{\Pi_{t+1}}{\bar{\Pi}} \frac{Y_{t+1}}{Y_t} + mc_t \mu_t + (1 - \mu_t) \quad (\text{A.23})$$

Symmetry and the fact that there is a unit continuum of final good producers and intermediate good producers also implies that $Y_{it} = Y_t = \theta_t K_{t-1}^\alpha H_t^{1-\alpha}$.

A.4 Capital Producers

Aggregate capital in the economy is the sum of entrepreneurial and household capital:

$$K_t = K_t^E + K_t^H. \quad (\text{A.24})$$

Perfectly competitive capital producers produce new capital by purchasing the final output good (this investment is denoted I) and combining it with undepreciated capital from last period, according to the following law of motion:

$$K_t = I_t + (1 - \delta) K_{t-1}. \quad (\text{A.25})$$

They face investment adjustment costs as in [Christiano et al. \(2005\)](#). Their period profits (in real terms) are:

$$\Omega_t^C = Q_t K_t - I_t - Q_t(1 - \delta)K_{t-1} - \frac{\kappa}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 I_t \quad (\text{A.26})$$

$$= I_t (Q_t - 1) - \frac{\kappa}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 I_t. \quad (\text{A.27})$$

Since they discount real payoffs using the household's real discount factor $\beta^t \lambda_t$, their maximization problem is:

$$\max_{I_t} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \lambda_t \Omega_t^C. \quad (\text{A.28})$$

And the first order condition simplifies to:

$$Q_t = 1 - \frac{\kappa}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 - \kappa \left(\frac{I_t}{I_{t-1}} - 1 \right) \frac{I_t}{I_{t-1}} + \Lambda_{t,t+1} \kappa \left(\frac{I_{t+1}}{I_t} - 1 \right) \left(\frac{I_{t+1}}{I_t} \right)^2. \quad (\text{A.29})$$

A.5 Market Clearing

Good market clearing implies that:

$$\begin{aligned} Y_t = & C_t + I_t + G_t + \frac{\theta_R}{2} \left(\frac{\Pi_t}{\bar{\Pi}} - 1 \right)^2 Y_t + \frac{\theta_W}{2} \left(\frac{\Pi_t^W}{\bar{\Pi}} - 1 \right)^2 w_t + \frac{\kappa}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 I_t \\ & + \Sigma_{ft} + \Sigma_{bt} + \frac{\kappa_H}{2} (K_t^H)^2 + \frac{C_{ft}}{\bar{\Pi}_t} + \frac{\kappa_D}{2} \left(\frac{R_{Dt}}{R_{Dt-1}} - 1 \right)^2 R_{Dt} D_t, \end{aligned} \quad (\text{A.30})$$

where Σ_{ft} and Σ_{bt} are default costs of entrepreneurial firms and banks, respectively. Real repossession costs of defaulting firms are given by

$$\Sigma_{ft} = \delta_M [Q_t(1 - \delta)K_{t-1}^E + z_t K_{t-1}^E] \int_0^\infty \int_0^{\bar{\omega}_{Ft}(\omega_k)} \omega_k \omega_j dF_{jt}(\omega_j) dF_{kt}(\omega_k), \quad (\text{A.31})$$

while the real repossession costs of defaulting banks are given by

$$\Sigma_{bt} = \frac{1}{\bar{\Pi}_t} \delta_B \left[\int_0^{\bar{\omega}_{kt}} \tilde{R}_{Lt}(\omega_k) L_{t-1} dF_{kt}(\omega_k) \right]. \quad (\text{A.32})$$

A.6 Proofs

A.6.1 Bank's Optimality Conditions

Deposits. Using the balance sheet constraint (26), deposit demand (27) and the default cutoff (23) in the bank's objective function (25) yields:

$$\mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \left[c_R L_{kt} + \int_{\bar{\omega}_{bkt+1}}^{\infty} \left((\tilde{R}_{Lkt+1}(\omega) - R_t) L_{kt} + \left[R_t - R_{Dkt} - \frac{\kappa_D}{2} \left(\frac{R_{Dkt}}{R_{Dkt-1}} - 1 \right)^2 R_{Dkt} \right] \left(\frac{R_{Dkt}}{R_{Dt}} \right)^{-\epsilon_D} D_t + (R_{t+1}^S - R_t) S_t^L + R_t \bar{B}_{kt} - C_{ft} \right) dF_{kt+1}(\omega) \right]. \quad (\text{A.33})$$

By the Leibniz rule, the derivative with respect to the bank's deposit rate R_{Dkt} is:

$$\mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \left[-1 + \epsilon_D - \epsilon_D (R_t - R_{Dkt}) - \kappa_D \left(\frac{R_{Dkt}}{R_{Dkt-1}} - 1 \right) \frac{R_{Dkt}}{R_{Dkt-1}} \right] (1 - F_{kt+1}(\bar{\omega}_{kt+1})). \quad (\text{A.34})$$

The term $\frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} [1 - F_{kt+1}(\bar{\omega}_{bkt+1})]$ can never be zero (unless the bank fails with certainty), hence the optimal deposit rate is characterized by:

$$-1 + \epsilon_D - \epsilon_D (R_t - R_{Dkt}) - \kappa_D \left(\frac{R_{Dkt}}{R_{Dkt-1}} - 1 \right) \frac{R_{Dkt}}{R_{Dkt-1}} = 0 \quad (\text{A.35})$$

which is independent of all other choice variables of the bank. $\square$

Loan Supply. It has just been shown that the optimal choice for the deposit rate R_{Dt}^* is independent of L_{kt} . Using this result, it is shown below that the bank's objective is convex in the loan volume for any given level of long-maturity bonds S_t^L (including the optimal level). Using the optimal deposit rate (31), the balance sheet constraint (26) and the default cutoff (23) in the bank's objective function (25), the latter can be stated as a function of the loan volume for any given level of S_t^L :

$$f(L_{kt}) = \mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \left[c_R L_{kt} + \int_{\bar{\omega}_{bkt+1}}^{\infty} \left((\tilde{R}_{Lkt+1}(\omega) - R_t) L_{kt} - (R_{Dt}^* + \frac{\kappa_D}{2} \left(\frac{R_{Dt}^*}{R_{Dt-1}^*} - 1 \right)^2 - R_t) D_t + (R_{t+1}^S - R_t) S_t^L + R_t \bar{B}_{kt} - c_f \right) dF_{kt+1}(\omega) \right]. \quad (\text{A.36})$$

By the Leibniz rule, we have:

$$f'(L_{kt}) = \mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \left[c_R + \int_{\bar{\omega}_{bkt+1}}^{\infty} \left[\tilde{R}_{Lkt+1}(\omega) - R_t \right] dF_{kt+1}(\omega) \right], \quad (\text{A.37})$$

and, as long as the bank's default threshold $\bar{\omega}_{bkt+1}$ exists (i.e. for a given amount of lending, there are states of the world in which the bank fails and other states of the world in which it is solvent):

$$f''(L_{kt}) = \mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \left[-\frac{\partial \bar{\omega}_{bkt+1}}{\partial L_{kt}} \left[\tilde{R}_{Lkt+1}(\bar{\omega}_{bkt+1}) - R_t \right] f_{kt+1}(\bar{\omega}_{bkt+1}) \right], \quad (\text{A.38})$$

where by definition of the default threshold ω_b and the implicit function theorem:

$$\frac{\partial \bar{\omega}_{bkt+1}}{\partial L_{kt}} = -\frac{\tilde{R}_L(\bar{\omega}_{bkt+1}) - R_t}{\frac{\partial \tilde{R}_L(\bar{\omega}_{bkt+1})}{\partial \omega_{bkt+1}} L_{kt}}. \quad (\text{A.39})$$

Before turning to the case in which the bank's default threshold exists (as will be the case in equilibrium), let us briefly discuss the case in which this threshold does not exist. First, the threshold may not exist because the bank makes losses from non-lending activities that exceed (with certainty) any possible payoff from lending (for a given loan rate, which is taken as given) for all levels of lending consistent with the capital requirement, i.e. $\forall L_{kt} \in [0, \bar{L}_{kt}]$. In this case, the bank fails with certainty and is obviously indifferent between any level of lending $L_{kt} \in [0, \bar{L}_{kt}]$. Second, the threshold may not exist because the payoff from non-lending activities is so high that it always exceeds any possible losses from lending $\forall L_{kt} \in [0, \bar{L}_{kt}]$. In this case, the lower bound of the integral in Eq. (A.37) becomes zero. The bank is then indifferent between any $L_{kt} \in [0, \bar{L}_{kt}]$ if:

$$\mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \left[c_R + \int_0^{\infty} \left[\tilde{R}_{Lkt+1}(\omega) - R_t \right] dF_{kt+1}(\omega) \right] = 0. \quad (\text{A.40})$$

and strictly prefers lending either $\bar{L}_{kt}$ (when the LHS of (A.40) is above zero) or 0 (when the LHS of (A.40) is below zero).

We now turn to the case in which the bank's default threshold exists. To shorten notation, let the realized return on physical capital for entrepreneurs (net of shocks ω) be denoted by

R_{Kt} :

$$R_{Kt} = \frac{\Pi_t[Q_t(1 - \delta) + z_t]}{Q_{t-1}}, \quad (\text{A.41})$$

and let the leverage of entrepreneurial firms be denoted:

$$\Theta_t = \frac{E_t}{Q_t K_t^E}. \quad (\text{A.42})$$

The bank does not internalize any effects of changes in the bank's default cutoff on the return on physical capital or firm leverage. Then by definition of the ex-post return on loans, we have:

$$\begin{aligned} \frac{\partial \tilde{R}_L(\bar{\omega}_{bkt+1})}{\partial \bar{\omega}_{bkt+1}} &= -\frac{\partial \bar{\omega}_{Ft+1}(\bar{\omega}_{bkt+1})}{\partial \bar{\omega}_{bkt+1}} R_{Lkt} f_{jt+1}(\bar{\omega}_{Ft+1}(\bar{\omega}_{bkt+1})) \\ &+ (1 - \mu_F) \frac{R_{Kt+1}}{1 - \Theta_t} \int_0^{\bar{\omega}_{Ft+1}(\bar{\omega}_{bkt+1})} \omega dF_{jt+1}(\omega) \\ &+ (1 - \mu_F) \bar{\omega}_{bkt+1} \frac{R_{Kt+1}}{1 - \Theta_t} \bar{\omega}_{Ft+1}(\bar{\omega}_{bkt+1}) f_{jt+1}(\bar{\omega}_{Ft+1}(\bar{\omega}_{bkt+1})) \quad (\text{A.43}) \\ &= [R_{Lkt} - (1 - \mu_F) R_{Lkt}] \left(-\frac{\partial \bar{\omega}_{Ft+1}(\bar{\omega}_{bkt+1})}{\partial \bar{\omega}_{bkt+1}} \right) + (1 - \mu_F) \frac{R_{Kt+1}}{1 - \Theta_t} \int_0^{\bar{\omega}_{Ft+1}(\bar{\omega}_{bkt+1})} \omega dF_{jt+1}(\omega) > 0. \quad (\text{A.44}) \end{aligned}$$

The second equation follows since by definition of $\bar{\omega}_{Ft+1}(\bar{\omega}_{bkt+1})$:

$$\frac{\partial \bar{\omega}_{Ft+1}(\bar{\omega}_{bkt+1})}{\partial \bar{\omega}_{bkt+1}} = -\frac{\bar{\omega}_{Ft+1}(\bar{\omega}_{bkt+1})}{\bar{\omega}_{bkt+1}} < 0. \quad (\text{A.45})$$

Hence $f''(L_{kt})$ simplifies to:

$$f''(L_{kt}) = \mathbb{E}_t \Lambda_{t,t+1} \frac{(\tilde{R}_L(\bar{\omega}_{bkt+1}) - R_t)^2}{\frac{\partial \tilde{R}_L(\bar{\omega}_{bkt+1})}{\partial \bar{\omega}_{bkt+1}} L_t} f_{kt+1}(\bar{\omega}_{bkt+1}) \geq 0. \quad (\text{A.46})$$

Summarizing the above discussion, it follows that:

$$f''(L_{kt}) \begin{cases} = 0 & \text{if } \nexists \bar{\omega}_{bkt+1} \text{ or } \tilde{R}_L(\bar{\omega}_{bkt+1}) = R_t, \\ > 0 & \text{else.} \end{cases} \quad (\text{A.47})$$

The objective function is thus convex in L_{kt} (but not strictly convex). Hence, when $f(L)$ is linear in $[0, \bar{L}_{kt}]$, the optimal lending decision is determined by the sign of $f'(L)$, which depends on the loan rate. Note that this implies that any equilibrium with $\tilde{R}_L(\bar{\omega}_{bkt+1}) = R_t$ involves $f'(L) > 0$ and hence $L_{kt} = \bar{L}_{kt}$. On the other hand, whenever $f(L)$ has a strictly convex region in $[0, \bar{L}_{kt}]$, the Karush-Kuhn-Tucker conditions characterize a *minimum* for a given loan rate.³³ Hence, the optimal loan supply follows a corner solution and either $L_{kt} = 0$ or $L_{kt} = \bar{L}_{kt}$ depending on the loan rate. $\square$

A.7 Contracting Problem Between Banks & Entrepreneurial Firms

Let λ_t^F denote the multiplier on the firm's financing constraint Eq. (7) and λ_t^{PC} the multiplier on the bank's participation constraint Eq. (36). Since the (atomistic) firm takes the total loan volume intermediated by the bank as given, the first order conditions of the contracting problem are given by:

$$(K_t^E) : \mathbb{E}_t \Lambda_{t,t+1}^E \int_0^\infty \int_{\bar{\omega}_{Ft+1}(\omega_k)}^\infty \omega_j \omega_k [Q_{t+1}(1 - \delta) + z_{t+1}] dF_k(\omega_j) dF_j(\omega_j) - Q_t \lambda_t^F - \lambda_t^{PC} \mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \left[\int_{\bar{\omega}_{bt+1}}^\infty \frac{\partial \tilde{R}_{Lt+1}(\omega)}{\partial K_t^E} L_t dF_k(\omega) \right] = 0 \quad (\text{A.48})$$

$$(L_{jt}) : \mathbb{E}_t \frac{\Lambda_{t,t+1}^E}{\Pi_{t+1}} \int_0^\infty \int_{\bar{\omega}_{Ft+1}(\omega_k)}^\infty (-R_{Lt}) dF_k(\omega_j) dF_j(\omega_j) + \lambda_t^F - \lambda_t^{PC} \mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \left[\int_{\bar{\omega}_{bt+1}}^\infty \frac{\partial \tilde{R}_{Lt+1}(\omega)}{\partial L_t} L_t dF_k(\omega) \right] = 0 \quad (\text{A.49})$$

$$(R_{Lt}) : \mathbb{E}_t \frac{\Lambda_{t,t+1}^E}{\Pi_{t+1}} \int_0^\infty \int_{\bar{\omega}_{Ft+1}(\omega_k)}^\infty (-L_t) dF_k(\omega_j) dF_j(\omega_j) = \lambda_t^{PC} \mathbb{E}_t \frac{\Lambda_{t,t+1}^B}{\Pi_{t+1}} \left[\int_{\bar{\omega}_{bt+1}}^\infty \frac{\partial \tilde{R}_{Lt+1}(\omega)}{\partial R_{Lt}} L_t dF_k(\omega) \right] \quad (\text{A.50})$$

³³To see this, note that the constraint $\gamma L_{kt} \leq \bar{L}_{kt}$ – where $\bar{L}_{kt} = \frac{\bar{B}_t}{\gamma}$ under fair-value capital requirements and $\bar{L}_{kt} = \frac{B_t^{BV}}{\gamma}$ under book-value capital requirements – is clearly linear in L_{kt} .

Using that $\bar{\omega}_{Ft+1}(\omega_k) = \frac{R_{Lt}L_j}{\omega_k \Pi_{t+1}[Q_{t+1}(1-\delta)K_{Et} + z_{t+1}K_{Et}]}$, these derivatives are given by:

$$\begin{aligned} \frac{\partial \tilde{R}_{Lt+1}(\omega_k)}{\partial K_{Et}} &= \frac{\omega_k(1-\delta_M)\Pi_{t+1}[Q_{t+1}(1-\delta) + z_{t+1}]}{L_t} \Phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) - \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \\ &+ \delta_M \frac{R_{Lt}}{K_{Et}} \frac{1}{\sigma_{jt+1}} \phi \left(\frac{\ln(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \end{aligned} \quad (\text{A.51})$$

$$\begin{aligned} \frac{\partial \tilde{R}_{Lt+1}(\omega_k)}{\partial L_t} &= -\frac{\omega_k(1-\delta_M)\Pi_{t+1}[Q_{t+1}(1-\delta) + z_{t+1}]K_{Et}}{L_t^2} \Phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) - \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \\ &- \delta_M \frac{R_{Lt}}{L_t} \frac{1}{\sigma_{jt+1}} \phi \left(\frac{\ln(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \end{aligned} \quad (\text{A.52})$$

$$\frac{\partial \tilde{R}_{Lt+1}(\omega_k)}{\partial R_{Lt}} = (1 - F_{kt+1}(\bar{\omega}_{Ft+1}(\omega_k))) - \delta_M \frac{1}{\sigma_{jt+1}} \phi \left(\frac{\ln(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \quad (\text{A.53})$$

A.7.1 Taylor Approximation

The expectation of the ex-post realized loan rate and its derivatives are highly non-linear functions of ω_k . Therefore, to solve the model in Dynare at third order, it is necessary to manually compute a third order approximation.³⁴ The procedure follows [Mendicino et al.](#)

³⁴Derivatives of external functions are currently only implemented in Dynare up to second order.

(2026). We need the approximation of just three terms involving ω_k :

$$A \equiv \Phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right), \quad (\text{A.54})$$

$$\frac{\partial A}{\partial \omega_k} = -\phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \frac{1}{\sigma_{jt+1}\omega_k}, \quad (\text{A.55})$$

$$\begin{aligned} \frac{\partial^2 A}{\partial \omega_k^2} = & -\phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \left(\frac{1}{\sigma_{jt+1}\omega_k} \right)^2 \\ & + \phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \frac{1}{\sigma_{jt+1}\omega_k^2}. \end{aligned} \quad (\text{A.56})$$

$$B \equiv \omega_k \Phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) - \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right), \quad (\text{A.57})$$

$$\frac{\partial B}{\partial \omega_k} = -\phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) - \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \frac{1}{\sigma_{jt+1}} + \Phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) - \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right), \quad (\text{A.58})$$

$$\begin{aligned} \frac{\partial^2 B}{\partial \omega_k^2} = & -\phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) - \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) - \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \frac{1}{\sigma_{jt+1}^2 \omega_k} \\ & - \phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) - \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \frac{1}{\sigma_{jt+1}\omega_k}. \end{aligned} \quad (\text{A.59})$$

$$C \equiv \frac{1}{\sigma_{jt+1}} \phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right), \quad (\text{A.60})$$

$$\frac{\partial C}{\partial \omega_k} = \frac{1}{\sigma_{jt+1}^2 \omega_k} \phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right), \quad (\text{A.61})$$

$$\begin{aligned} \frac{\partial^2 C}{\partial \omega_k^2} = & -\frac{1}{\sigma_{jt+1}^2 \omega_k^2} \phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \\ & + \frac{1}{\sigma_{jt+1}^3 \omega_k^2} \phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right) \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right)^2 \\ & - \frac{1}{\sigma_{jt+1}^3 \omega_k^2} \phi \left(\frac{\log(\bar{\omega}_{Ft+1}(\omega_k)) + \frac{\sigma_{jt+1}^2}{2}}{\sigma_{jt+1}} \right). \end{aligned} \quad (\text{A.62})$$

Then, the expected ex-post realized loan rate can be approximated as:

$$\mathbb{E}_t \tilde{R}_{t+1}^L \approx \sum_{i=1}^N \left(\int_{x_i}^{x_{i+1}} T \left(\tilde{R}_{t+1}^L \right) (\omega_k) dF_{kt+1}(\omega_k) \right) + [1 - F_{kt+1}(x_{N+1})] R_t^L. \quad (\text{A.63})$$

where the Taylor Approximation of the ex-post realized loan rate around a point $\bar{x}_i = \frac{x_i + x_{i+1}}{2}$ is given by:

$$T \left(\tilde{R}_{t+1}^L \right) (\omega_k) = \tilde{R}_{t+1}^L(\bar{x}_i) + \frac{\partial \tilde{R}_{t+1}^L}{\partial \omega_k} (\omega_k - \bar{x}_i) + \frac{1}{2} \frac{\partial^2 \tilde{R}_{t+1}^L}{\partial \omega_k^2} (\omega_k - \bar{x}_i)^2. \quad (\text{A.64})$$

Using the expressions just derived:

$$T \left(\tilde{R}_t^L \right) (\omega_k) = \frac{(1 - \delta_M) \Pi_{t+1} [Q_{t+1}(1 - \delta) + z_{t+1}] K_t^E}{L_t} T(B) + R_t^L (1 - T(A)). \quad (\text{A.65})$$

Then:

$$\begin{aligned} \int_{x_i}^{x_{i+1}} T \left(\tilde{R}_{t+1}^L \right) (\omega_k) dF_{kt+1}(\omega_k) &= Q_0(\bar{x}_i) + Q_1(\bar{x}_i) \int_{x_i}^{x_{i+1}} \omega_k dF_{kt+1}(\omega_k) \\ &\quad + Q_2(\bar{x}_i) \int_{x_i}^{x_{i+1}} \omega_k^2 dF_{kt+1}(\omega_k), \end{aligned} \quad (\text{A.66})$$

where

$$Q_0(\bar{x}_i) = [F_{kt+1}(\omega_{i+1}) - F_{kt+1}(\omega_i)] \left[\tilde{R}_{t+1}^L(\bar{x}_k) - \bar{x}_i \frac{\partial \tilde{R}_{t+1}^L}{\partial \omega_k} + \frac{1}{2} \bar{x}_i^2 \frac{\partial^2 \tilde{R}_{t+1}^L}{\partial \omega_k^2} \right], \quad (\text{A.67})$$

$$Q_1(\bar{x}_i) = \left[\frac{\partial \tilde{R}_{t+1}^L}{\partial \omega_k} - \bar{x}_i \frac{\partial^2 \tilde{R}_{t+1}^L}{\partial \omega_k^2} \right], \quad (\text{A.68})$$

$$Q_2(\bar{x}_i) = \frac{1}{2} \frac{\partial^2 \tilde{R}_{t+1}^L}{\partial \omega_k^2}. \quad (\text{A.69})$$

We proceed similarly for the derivatives of the ex-post loan rate.

B Calibration

This Appendix presents additional details on the calibration strategy, and provides full details of the data sources.

where $\epsilon_{mw} \sim N(0, 1)$. The length of the simulation is 1700 quarters after a burn-in of 1500 periods. The burn-in period ensures that the ergodic distribution is reached.³⁵ We specify priors on some parameters due to SMM’s known tendency to pick parameters not supported by micro data (see [An and Schorfheide, 2007](#) and [Ruge-Murcia, 2012](#)).³⁶

Parameter	Description	Prior	Variance
ϕ_{Π}	Taylor Rule Weight Inflation	1.5	6
ϕ_Y	Taylor Rule Weight Inflation	0.5	2
ϕ_R	Taylor Rule Smoothing	1.5	2
κ	Investment Adjustment Costs	7	40

Table 5: Priors

B.1 Data Sources

Gross Domestic Product (GDP) deflator OECD Quarterly National Accounts. Euro Area (20 countries). Deflator, OECD reference year, seasonally adjusted. Millions of Euro. Index.

Population OECD Quarterly National Accounts. Euro Area (20 countries). Total Population. Thousands.

GDP OECD Quarterly National Accounts. Euro Area (20 countries). Gross domestic product - expenditure approach. National currency, current prices, quarterly levels, seasonally adjusted. Millions of Euro. *Transformation*: Divided by GDP deflator and total population.

³⁵We follow [Born and Pfeifer \(2014\)](#) who use the same burn-in length.

³⁶As discussed in [Born and Pfeifer \(2014\)](#), these priors are relatively flat.

Consumption Private final consumption expenditure. OECD Quarterly National Accounts. Euro Area (20 countries). National currency, current prices, quarterly levels, seasonally adjusted. Millions of Euro. *Transformation*: Divided by GDP deflator and total population.

Investment OECD Quarterly National Accounts. Euro Area (20 countries). Gross fixed capital formation. National currency, current prices, quarterly levels, seasonally adjusted. Millions of Euro. *Transformation*: Divided by GDP deflator and total population.

Inflation Log-change in GDP Deflator.

Employment OECD Quarterly National Accounts. Euro Area (20 countries). Employment, total (Persons). Thousands.

Hours OECD Quarterly National Accounts. Euro Area (20 countries). Employment, total (Hours Worked). Millions. *Transformation*: Demeaned Hours/Employment ($H = 1$ in the model corresponds to average hours worked).

Wages OECD Quarterly National Accounts. Euro Area (20 countries). Wages, total. Millions. *Transformation*: Wages divided by total hours worked and GDP deflator.

Share of Capital Managed by Households [Mendicino, Nikolov, Suarez, and Supera \(2020\)](#)

Rate of return on equity (RoE), bank default probability, and firm default probability [Mendicino et al. \(2026\)](#).

Loan-to-bond ratio [Hoffmann et al. \(2019\)](#)

Loans ECB Statistical Data Warehouse. The ECB does not report values for a hypothetical constant composition Euro Area, such that care must be taken to compute relationships with the measure of GDP for the corresponding countries. For this reason, we construct data

for the following countries: Austria, Belgium, Germany, Spain, Finland, France, Greece, Ireland, Italy, Luxembourg, Netherlands and Portugal.

The ECB data we use is only available monthly, so for each of the variables below, population weighted averages are computed for each quarter. Let $\omega_{ct} = \frac{Population_{ct}}{\sum_c Population_{ct}}$, where countries are indexed by c , quarters by t and month by m . ω_c are computed from the same OECD population data used above. Then the population-weighted averages for each quarter of variable x are:

$$\bar{x}_t = \frac{1}{4} \sum_{m \in t} \sum_c \omega_c x_{cm} \quad (\text{A.70})$$

(L1) : Loans vis-a-vis euro area NFCs reported by MFIs excl. ESCB. Up to 1 year maturity. Euro area (changing composition) counterpart, Non-Financial corporations (S.11) sector, denominated in Euro. Outstanding amounts at the end of the period (stocks). Monthly.

(L2) : Loans vis-a-vis euro area NFCs reported by MFIs excl. ESCB. Over 1 and up to 5 years maturity. Euro area (changing composition) counterpart, Non-Financial corporations (S.11) sector, denominated in Euro. Outstanding amounts at the end of the period (stocks). Monthly.

(L3) : Loans vis-a-vis euro area NFCs reported by MFIs excl. ESCB. Over 5 years maturity. Euro area (changing composition) counterpart, Non-Financial corporations (S.11) sector, denominated in Euro. Outstanding amounts at the end of the period (stocks). Monthly.

Sum of: (L1) + (L2) + (L3) of all countries listed above.

Transformation: Population weighted average for each quarters. Seasonally adjusted with X-13 ARIMA using the R package seasonal. Divided by nominal GDP and multiplied by real per capita GDP to get real per-capita loans. To compute the average loan-to-GDP ratio the sum of real per capita GDP for the same countries is used.

Loan rate spread ECB Statistical Data Warehouse.

(LR1): Annualised agreed rate (AAR) / Narrowly defined effective rate (NDER), Credit and other institutions (MFI except MMFs and central banks) reporting sector - Loans, Up to 1 year original maturity, Outstanding amount business coverage, Non-Financial corporations (S.11) sector, denominated in Euro. Monthly.

(SR1): Euro Interbank Offered Rate.

(LR2): Annualised agreed rate (AAR) / Narrowly defined effective rate (NDER), Credit and

other institutions (MFI except MMFs and central banks) reporting sector - Loans, Over 1 and up to 5 years original maturity, Outstanding amount business coverage, Non-Financial corporations (S.11) sector, denominated in Euro. Monthly.

(SR2): Yield curve spot rate, 2 year maturity. Government bond, nominal, all issuers whose rating is triple A - Euro area (changing composition). Monthly.

(LR3): Annualised agreed rate (AAR) / Narrowly defined effective rate (NDER), Credit and other institutions (MFI except MMFs and central banks) reporting sector - Loans, Over 5 years original maturity, Outstanding amount business coverage, Non-Financial corporations (S.11) sector, denominated in Euro. Monthly.

(SR3): Yield curve spot rate, 5 year maturity. Government bond, nominal, all issuers whose rating is triple A - Euro area (changing composition). Monthly.

Transformation:

$$\frac{((LR1) - (SR1))(L1) + ((LR2) - (SR2))(L2) + ((LR3) - (SR3))(L3)}{(L1) + (L2) + (L3)}$$

of all countries listed above. Compute population weighted average for each quarters.

Safe Rate ECB Statistical Data Warehouse. Using the data just described, we approximate the safe rate (the data counterpart to R_t) as:

$$\frac{(SR1)(L1) + (SR2)(L2) + (SR3)(L3)}{(L1) + (L2) + (L3)}$$

of all countries listed above. Then we compute the population weighted average for each quarter.

C Details Computation Consumption Equivalents

Denote variable x in model a by x_t^a and in model b by x_t^b , where a and b will be specified below. To compute consumption equivalents (CE) Δ_{CE} , we first calculate the expected

lifetime utility at time t under two different models:³⁷

$$\mathbb{E}(U_t^a) = \mathbb{E} \left(\mathbb{E}_t \sum_{n=0}^{\infty} \beta^n \left[\frac{\{C_{t+n}^a(h) - bC_{t+n-1}^a(h)\}^{1-\sigma}}{1-\sigma} - \frac{\xi_H H_{t+n}^a(h)^{1+\varphi_H}}{1+\varphi_H} \right] \right) \quad (\text{A.71})$$

and similarly for model b . Next, the expected utility in model b can be related to the utility in model a as follows:

$$\mathbb{E}(U_t^a) = \mathbb{E} \left(\mathbb{E}_t \sum_{n=0}^{\infty} \beta^n \left[\frac{\{(1 + \Delta_{CE})(C_{t+n}^b(h) - bC_{t+n-1}^b(h))\}^{1-\sigma}}{1-\sigma} - \frac{\xi_H H_{t+n}^b(h)^{1+\varphi_H}}{1+\varphi_H} \right] \right) \quad (\text{A.72})$$

Under log-utility ($\sigma = 1$), using Eq. (A.71) in Eq. (A.72) yields:

$$\mathbb{E}(U_t^a) = \frac{\ln(1 + \Delta_{CE})}{1 - \beta} + \mathbb{E}(U_t^b) \quad (\text{A.73})$$

It follows:

$$\Delta_{CE} = \exp[(1 - \beta)(\mathbb{E}\{U_t^a - U_t^b\})] - 1 \quad (\text{A.74})$$

Model a is for example of the model under fair value requirements, while model b is the model under book-value requirements. The interpretation is that households, could they choose, would require $100\Delta_{CE}\%$ of consumption in every period to remain in the economy with book-value requirements.

³⁷This may be interpreted as the expected utility of a person born at an arbitrary time t in the respective economy.

D Additional Tables and Figures

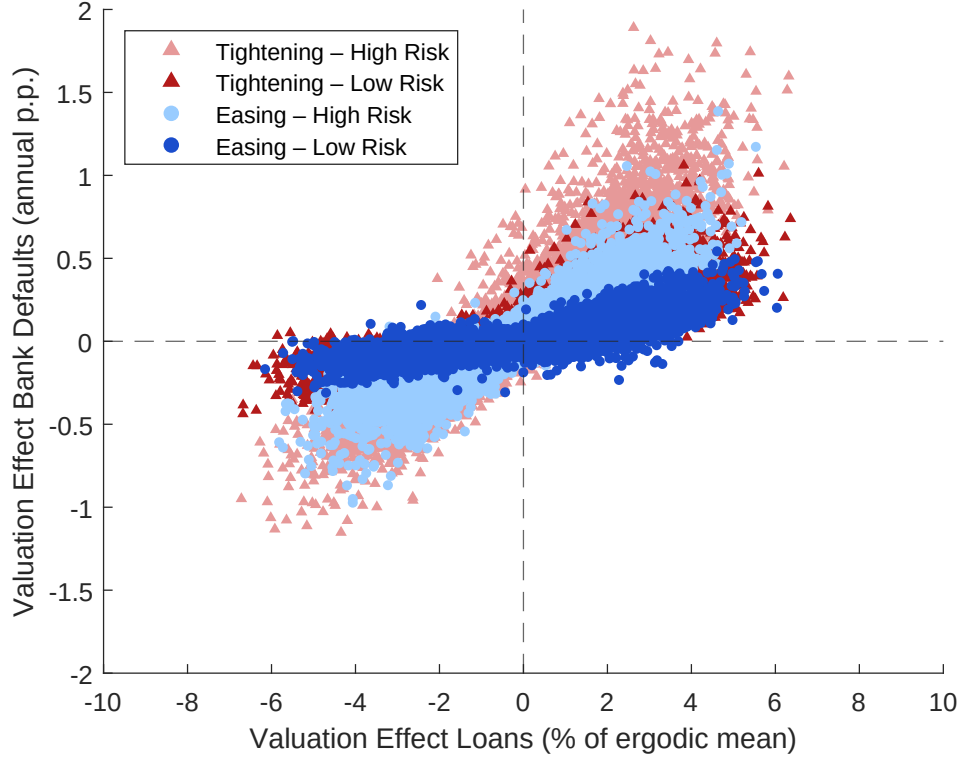


Figure 8: *Valuation Effects on Lending and Bank Defaults: Correlation by Regime*

Notes: Tightening episodes are defined as those in which the real policy rate is above the one implied by the Taylor Rule, i.e. $\tau_t > 1$, and easing episodes as those in which $\tau_t < 1$. Stability episodes are those in which the variance of the island-specific shock σ_{ω_k} is particularly low (below its median), and instability episodes those with σ_{ω_k} above its 75th percentile. The Valuation Effect on Loans is defined as the difference in bank loans between the baseline economy (book-value regime) and the alternate economy (fair-value regime) for different levels of unrealized losses or gains. The difference in bank loans is reported as a percentage of the ergodic mean of loans under the book-value regime regime. The Valuation Effect on Bank Defaults is defined as the difference in the annualized bank default probability between the baseline economy (book-value regime regime) and the alternate economy (fair-value regime) for different levels of unrealized losses or gains.

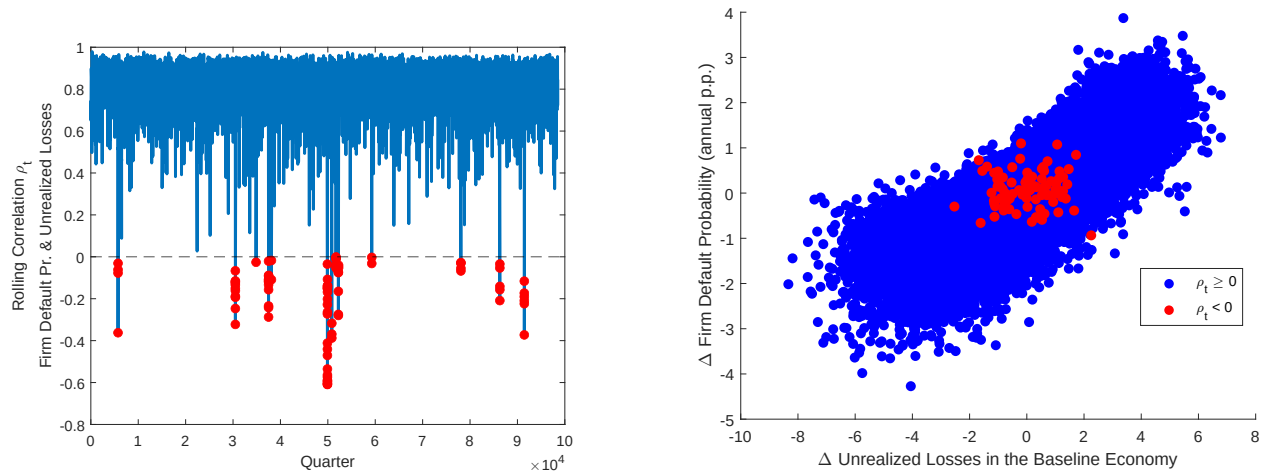


Figure 9: *Correlation Interest and Credit Risk*

Notes: The left panel plots the time series of the rolling correlation between the change in unrealized losses and the change in the firm default probability in the baseline economy in a window of 12 quarters. The right panel depicts a scatter plot of the same variable, colored by the sign of the rolling correlation.

Table 6: Parameter Values

Symbol	Value	Description
α	0.25	Capital Share Production
β	0.995	HH Discount Factor
δ	0.025	Capital Depreciation Rate
θ_R	68.4	Rotemberg Price Adjustment Cost
θ_W	1.13e+03	Rotemberg Wage Adjustment Cost
φ_H	1	Inverse Frisch Elasticity of Labor
μ	7.25	Elasticity of Substitution Final Goods
ϵ_W	5	Elasticity of Substitution Labor
κ	7.85	Investment Adjustment Cost
κ_H	0.00831	Capital Management Cost
σ	1	HH Risk Aversion (Consumption)
ξ_H	0.835	Disutility of Labor
θ_B	0.965	Survival Bankers
θ_E	0.969	Survival Entrepreneurs
γ	0.08	Regulatory Capital Requirement
ϕ_Π	1.07	Taylor Rule Weight Inflation
ϕ_y	0.00115	Taylor Rule Weight Output
ϕ_R	0.648	Taylor Rule Weight Smoothing Weight
ξ_B	0.5	Endowment Bankers
ξ_E	8.41e-07	Endowment Entrepreneurs
b	0.743	Habit Consumption
δ_M	0.3	Loss Entrepreneurial Default
δ_B	0.3	Loss Bank Default
$\bar{\sigma}_{\omega_j}$	0.133	Steady State Std Idiosyncratic Shock
$\bar{\sigma}_{\omega_k}$	0.112	Steady State Std Island Shock
σ_{ω_j}	0.000244	Std Idiosyncratic Risk Shock
σ_{ω_k}	0.0137	Std Island Risk Shock
ρ_{ω_j}	0.655	Autocorr. Idiosyncratic Risk Shock
ρ_{ω_k}	0.996	Autocorr. Island Risk Shock
ρ_θ	0.0158	Autocorr. Productivity Shock
ρ_p	0.607	Autocorr. Cost Push Shock
ρ_g	0.83	Autocorr. Gov. Spending Shock
ρ_τ	0.961	Autocorr. Monetary Policy Shock
σ_θ	0.000254	Std. Productivity Shock
σ_p	0.128	Std. Cost Push Shock
σ_g	0.00382	Std. Gov. Spending Shock
σ_τ	0.000281	Std. Monetary Policy Shock
S^L	1.09	Real Supply Central Bank Asset
m	13.6	Gov. Bond Maturity
ϕ_{lc}	0	Bank Liquidity Management Cost
c_f	0.00491	Bank Fixed Cost
c_R	0.00274	Bank Relationship Lending Benefit
ϵ_D	-400	Deposits Elasticity of Substitution
κ_D	1184.77	Deposits Adjustment Cost